Storage, drought and climate change: the reliability of the canal supply from Klein Maricopoort Dam and the role of groundwater in A31E, North West Province

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Storage, drought and climate change: the reliability of the canal supply from Klein Maricopoort Dam and the role of groundwater in A31E, North West Province

Abstract

Irrigation in quaternary catchment A31E, North West Province, depends on a canal from Klein Maricopoort Dam, which carried less than the registered volume of 2.34 Mm³/a in thirteen successive years from 1984. This desktop study assessed how reliable the canal supply is, how far groundwater within A31E could offset its shortfalls and how both respond to a drier climate, and it screened the floods and contaminant paths along the irrigated reach. A monthly water balance of the dam (1936–2009) was run with and without a restriction rule, and a steady-state groundwater model was run in two structures that differ in the depth of outflow. With the rule, the canal delivers the registered volume in 45% of the years, and without restriction in 77%. In one structure, pumped groundwater would come mostly from the river that feeds Kromellenboog Dam. In the sensitivity cases for about 2050, the reliability of the canal is lower, most in the dry case. The 1:100-year flood, estimated with the SDF method, covers about a quarter of the irrigated land in the strip below the dam, and the modelled reach reduces its peak by only 1% to 2%. The river below the dam is above the salinity target for irrigation in most samples. The registered volume cannot be assured by the dam, and groundwater cannot yet be relied on as a backup, because its effect on the river has not been established.

Keywords: canal supply reliability; reservoir water balance; multi-year drought; groundwater flow model; capture; climate change; design flood; particle tracking

1. Introduction

In semi-arid regions, irrigated agriculture depends largely on water that has been stored. Irrigators in a semi-arid South African catchment have coped with past droughts mainly through the construction of water storage and the adoption of more efficient irrigation (Lankford et al., 2023). The security of an irrigation supply depends on how reliably a dam can deliver water, which is described by how often the supply fails, how quickly it recovers and how severe a failure is (Hashimoto et al., 1982). In the Marico area of the North West Province, irrigation was the largest water use in 2000, at approximately 32 million cubic metres per year (Mm^3/a), and most of this water was drawn from surface water below dams such as Marico-Bosveld and Klein Maricopoort (DWAF, 2004:ii, 52).

In South Africa, the quaternary catchment serves as the fundamental hydrological unit for managing water resources (DWS, s.a.d). In semi-arid quaternary catchments, river flow and groundwater recharge do not respond to rainfall in simple proportion. Where about 500 mm of rain falls per year, a decrease of 10% in rainfall may reduce perennial surface drainage by half (De Wit & Stankiewicz, 2006), and groundwater in drylands is recharged episodically, often through focused recharge from ephemeral flows after intense rain (Cuthbert et al., 2019). Dry years tend to occur in runs, and six of the eight most severe two-year droughts in southern Africa between 1901 and 1999 ended between 1983 and 1996 (Rouault & Richard, 2005). During such dry periods, groundwater sustains river flow as baseflow, especially in dry seasons (Xie et al., 2024). Consequently, water availability in a quaternary catchment is strongly influenced by the sequence of wet and dry years and by the interaction between surface water and groundwater.

Groundwater is a crucial water resource in many semi-arid regions, because it buffers the effects of rainfall variability (Calow et al., 2010). In South Africa, groundwater contributes approximately 15% of the water consumed nationally (DWAF, cited by Knüppe, 2011:69), but in the Marico area it supplied 6 of the 32 Mm^3/a used for irrigation (DWAF, 2004:52). Over-abstraction has already lowered groundwater levels in the dolomite aquifers of the North West Province (Cobbing, 2018). Its use for irrigation is further limited by the yield of boreholes, since yields above 5 L/s are not widespread in Africa (MacDonald et al., 2012).

The A31E quaternary catchment is a semi-arid catchment in which irrigation depends on stored surface water. The largest part of its registered water use is not groundwater, but water that appears to be supplied by a canal from Klein Maricopoort Dam, which stands on the upstream boundary of the catchment (DWAF, 2004:50; DWS, s.a.f). A failure of this supply directly affects the water users in the catchment, and it raises the question of whether groundwater within A31E could sustain them through such a failure.

1.1 Research problem

Klein Maricopoort Dam stores water that is generated in the catchments above A31E. Of the 3.38 Mm^3/a of water use registered in A31E, 2.34 Mm^3/a (69%) is registered as scheme use on 442 ha within about 15 km of the dam wall, while 0.285 Mm^3/a is registered from boreholes

(DWS, s.a.f). These scheme records are inferred to represent the users of the canal, because the Department lists a canal scheme for the dam and the land mapped as irrigated lies below it (DFFE, 2024; DWS, 2024).

The records show that this supply has failed for extended periods. Between October 1983 and September 1995, the dam stood below 25% of its capacity in 139 of the 144 months, and this period is referred to as the drought throughout this study. During these years the canal carried a mean of 0.36 Mm³/a, compared with a median of 2.02 Mm³/a over 72 years of record (DWA, 2012; DWS, s.a.b). The inflow to the dam did not recover when the rainfall did: it stayed below its long-term median from 1982 to 1994, although the rainfall of 1986–1995 averaged 99% of its long-term mean (Bailey & Pitman, 2016). The Department reports a yield of 1.4 Mm³/a for the dam, without a stated method and with very low confidence in the hydrological information of the Marico (DWAF, 2004:50, 83). How often, for how long and by how much the deliveries of the canal fall below the registered volume is not known.

It is equally uncertain whether groundwater within A31E could serve as an alternative source. The estimated recharge of 17.336 Mm³/a is far larger than the current groundwater use (DWS, 2021). The mapped borehole yields are low, each borehole has a single recorded water level, and no pumping tests or water-level series exist (DWS, s.a.c). The only gauged stream inside A31E was dry in 42 of the 50 months from February 1984 (DWS, s.a.b), which suggests that groundwater storage declined during the same drought that emptied the dam.

Climate change adds a further uncertainty. Warming is projected with considerably more confidence than any change in rainfall (DEA, 2013), and droughts are projected to become more frequent and longer (Trisos et al., 2022). For 2041–2060, the climate models assessed by the IPCC project a warming of about 1.5 to 2.9 °C for eastern southern Africa (ESAF), the region in which A31E lies, and a change in annual rainfall of about −8.5% to +3%, with a median of about −3% (10th to 90th percentiles; calculated from Gutiérrez et al., 2021; Iturbide et al., 2020). The problem addressed in this study is that the reliability of the canal supply, the extent to which groundwater could offset its shortfalls and the sensitivity of both to climate change have not been established. Water users and the responsible authorities lack a quantified basis for drought planning.

1.2 Aim, objectives and research questions

The aim of this study is to quantify the reliability of the canal supply from Klein Maricopoort Dam, the extent to which groundwater within A31E could offset its shortfalls and the response of both to climate change, so that drought planning in the catchment can rest on the records rather than on a registered volume. The study is a desktop assessment of existing records, intended for the Department of Water and Sanitation (DWS), the North West Premier's Office and the consulting partner AGES. Flood risk along one reach of the Klein Marico River and the pathways along which contaminants could move are screened, and the transport of contaminants is not modelled.

The objectives of the study are:

1. To compile and review the existing hydrological, geohydrological and water-use information for A31E, identify its gaps, and develop a conceptual model of how water moves through the catchment.
2. To simulate the water balance of Klein Maricopoort Dam and its canal and quantify the reliability of supply against the registered scheme volume.
3. To assess with a groundwater flow model how far abstraction within A31E could offset shortfalls in the canal supply, and the limits that the data place on this assessment.
4. To assess the effects of climate change by about 2050 on the reliability of the canal supply and on groundwater.
5. To screen the flood risk along the irrigated reach of the Klein Marico River and the pathways along which contaminants from the land could reach the rivers.
6. To provide evidence-based recommendations, including priorities for data collection, to the DWS, the North West Premier's Office and AGES.

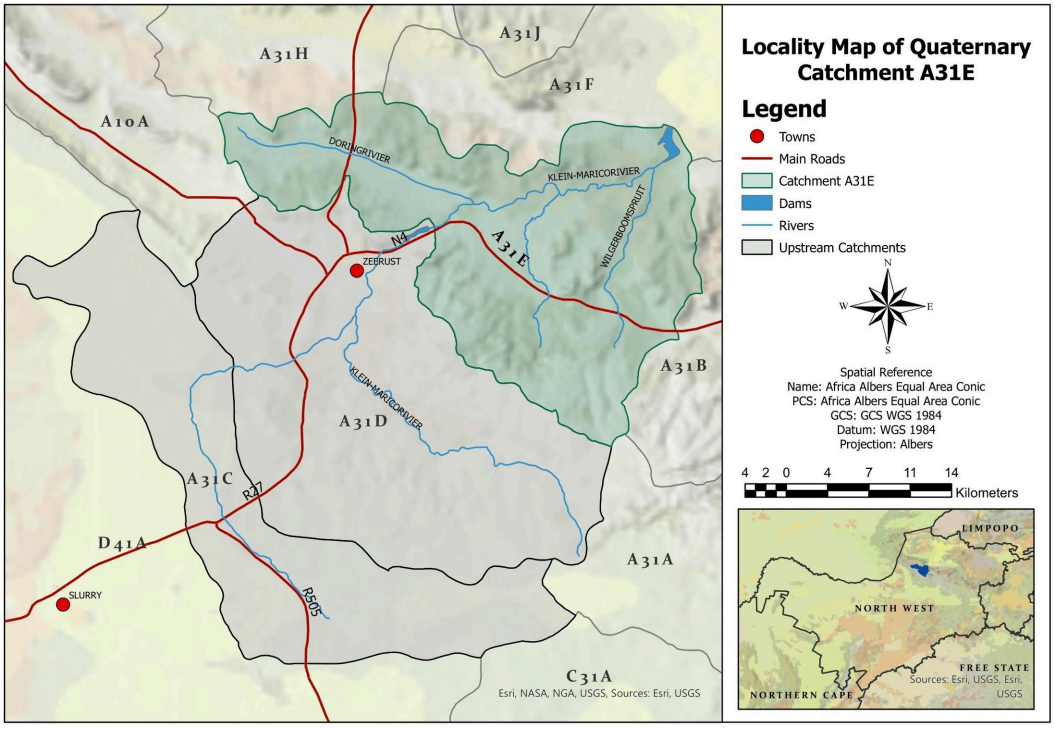
The research questions are:

1. How reliable is the canal supply from Klein Maricopoort Dam to irrigators in quaternary catchment A31E under the historical sequence of wet and dry years, to what extent could groundwater within A31E offset the shortfalls that arise when the dam is drawn down, and how would both change under plausible climate conditions for about 2050?
2. How large are the floods along the irrigated reach, and along which paths could contaminants from the land reach the receptors?

2. Study Area

2.1 Location and the two dams

Quaternary catchment A31E lies in the North West Province, east of Zeerust, and covers approximately 602 km² (Bailey & Pitman, 2016). It drains to the Marico River, part of the upper Limpopo River basin (Jury, 2016), within the Limpopo-Olifants Water Management Area (DWS, 2023). The Klein Marico River enters A31E at the wall of Klein Maricopoort Dam and leaves the catchment through Kromellenboog Dam (Map 1). The reservoir of the upper dam and its whole drainage area lie outside the catchment, and none of the runoff generated within A31E can reach Klein Maricopoort Dam. The water stored in the upper dam comes entirely from outside A31E, while the runoff of A31E flows to Kromellenboog Dam, which is operated for an irrigation scheme in the downstream catchment A31F (DWAF, 2004:50, 57).



Map 1: Locality of quaternary catchment A31E, showing the Klein Marico River, Klein Maricopoort Dam at the upstream boundary, Kromellenboog Dam at the outlet, the upstream catchments A31D and A31C, Zeerust and the main roads. Inset: position in the North West Province.

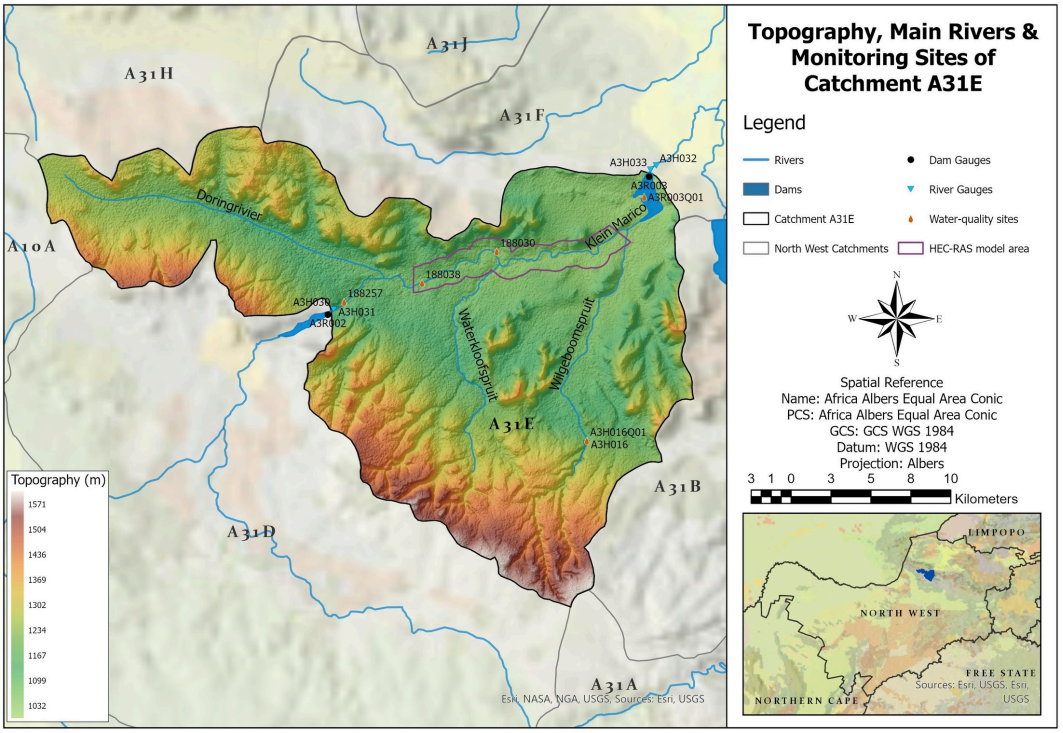
The canal leaves the upper dam at its wall, and the Department lists an irrigation scheme for the dam that is supplied “from the canal system” (DWS, 2024). The dam also releases water to the river. It was raised in 1966, and its survey of 1982 gives a capacity of 7.07 Mm³ at full supply (DWA, 2012).

2.2 Climate

The climate of the Marico area is semi-arid, with rainfall that is “strongly seasonal with rainfall occurring as thunderstorms in summer” (DWAF, 2004:18). The mean annual precipitation (MAP) of A31E is 597 mm in the Water Resources of South Africa, 2012 study (WR2012), which takes its rainfall from the series of rainfall zone A3B (Bailey & Pitman, 2016:14). The mean annual evaporation of the evaporation zone of A31E is 1 901 mm (Bailey & Pitman, 2016).

2.3 Topography and drainage

A31E falls from the high ground in the south, where the Waterkloofspruit and the Wilgeboomspruit rise, to Kromellenboog Dam in the north-east. A valley crosses the northern part of the catchment from the Doringrivier in the west to the dam (Map 2). Four streams drain A31E. The Klein Marico River flows from the wall of Klein Maricopoort Dam through the valley to Kromellenboog Dam. The Doringrivier drains the western arm and joins it below the dam wall, the Waterkloofspruit joins it further downstream, and the Wilgeboomspruit reaches it at the head of Kromellenboog Dam.

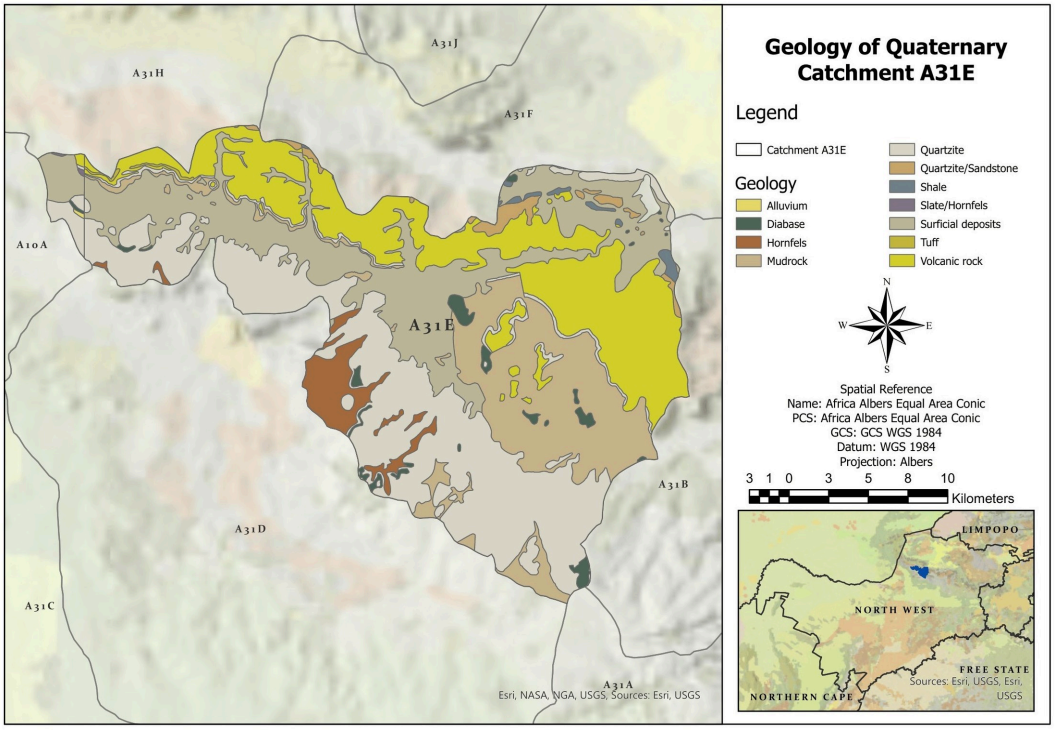


Map 2: Topography, main rivers and monitoring sites of A31E, with the area of the HEC-RAS model along the Klein Marico River. The stations at the wall of Klein Maricopoort Dam lie within 400 m of one another, and only some are labelled.

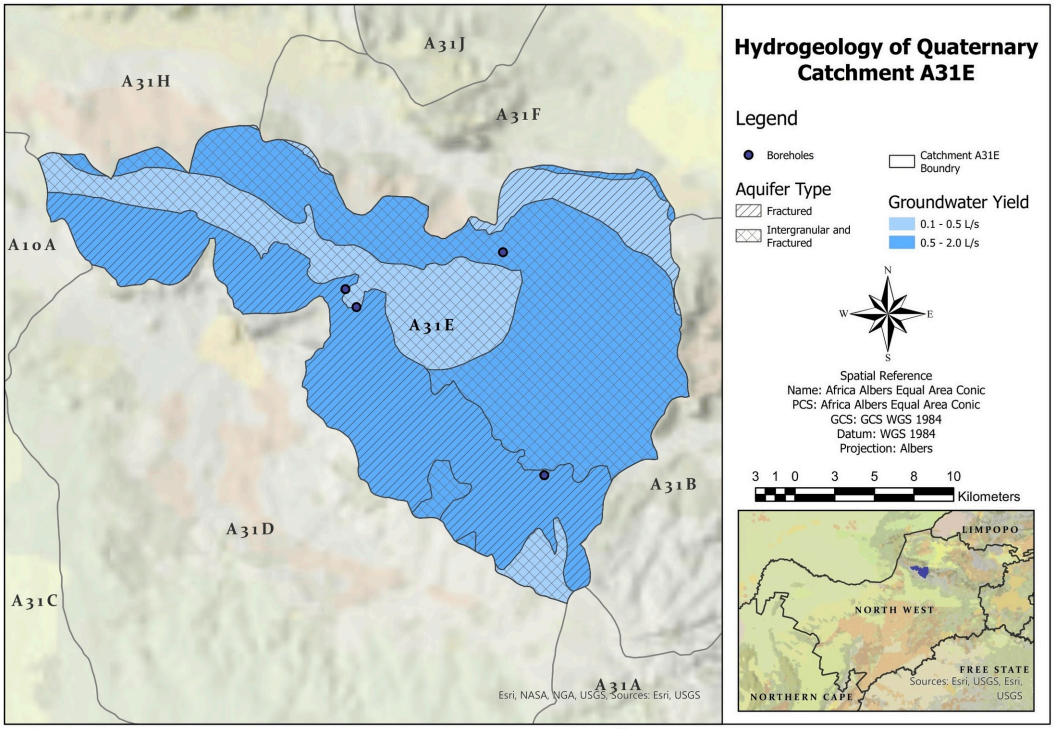
2.4 Geology and hydrogeology

A31E is underlain by rocks of the Pretoria Group, which are cut by diabase and partly covered by surface deposits (Map 3; Michaluk & Keyser, 1991; Walraven, 1981). Quartzite and sandstone, mainly of the Timeball Hill Formation, form the high ground in the south. The andesitic lava of the Hekpoort Formation lies in the east, mudrock in the centre-east and hornfels south-west of the centre, and diabase occurs in small bodies. Surface deposits, including

alluvium, lie mainly along the valley in the north. Groundwater occurrence depends primarily on geology, weathering and effective rainfall (MacDonald et al., 2012:2). On the hydrogeological map, A31E is classified as an intergranular and fractured aquifer and a fractured aquifer, and karst aquifers do not occur in A31E. The mapped median yield of boreholes is 0.5–2.0 L/s over most of A31E and 0.1–0.5 L/s in the rest (Map 4; Barnard, 1999). The National Groundwater Archive (NGA) holds the borehole records of the catchment (DWS, s.a.c). The Department judged that groundwater in the Marico can meet domestic supplies "but not bulk water requirements for purposes such as irrigated agriculture" (DWAF, 2004:55–56).



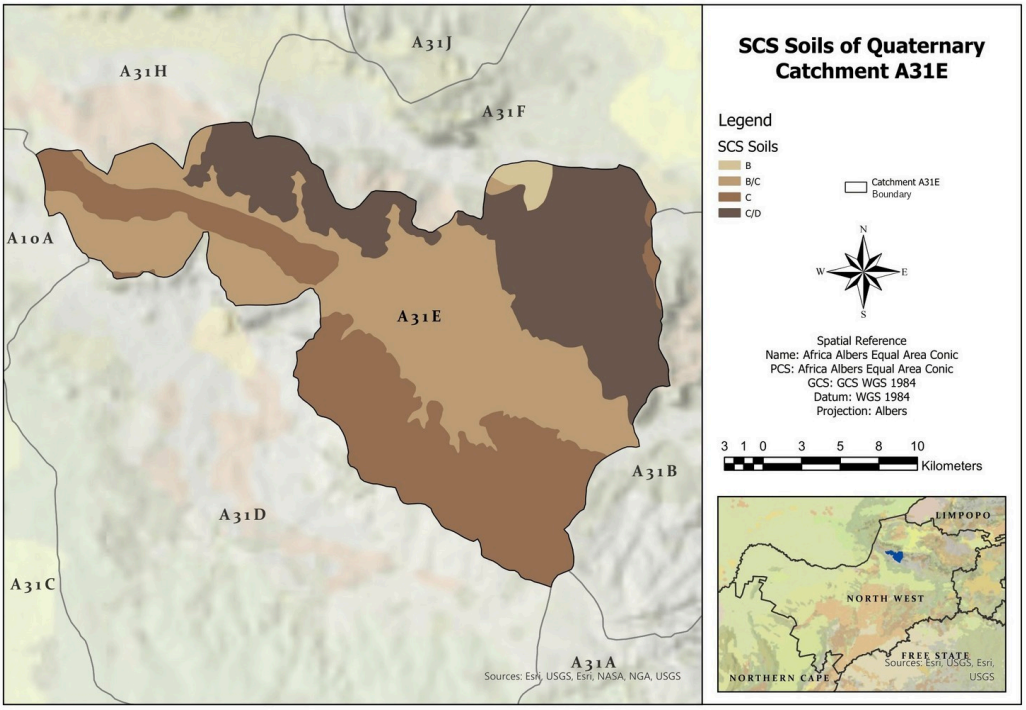
Map 3: Geology of A31E, showing the rock types of the 1:250 000 geological maps (Michaluk & Keyser, 1991; Walraven, 1981). The rock types are combined into four groups: quartzite and sandstone; volcanic rocks, diabase and tuff; mudrock, shale and hornfels; and surface deposits, which include the alluvium.



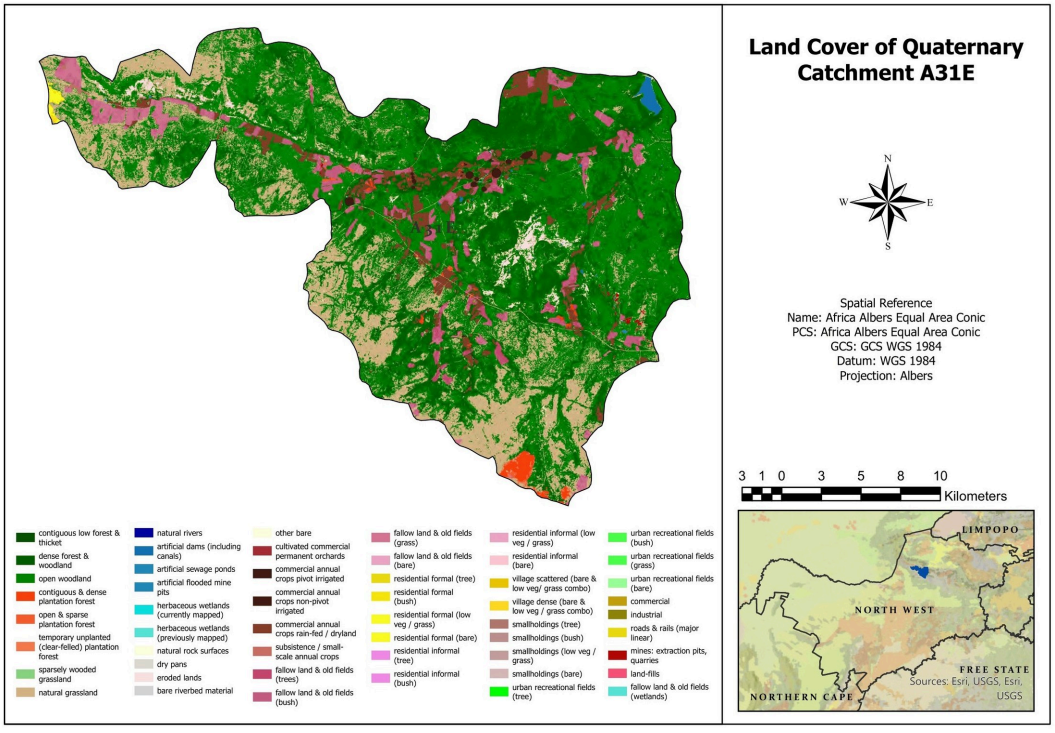
Map 4: Aquifer types and mapped borehole-yield classes in A31E (Barnard, 1999), with the four boreholes that have a static water level (DWS, s.a.c). The yield classes are mapped medians, not measured yields.

2.5 Soils and land cover

The soils of A31E are assigned to the hydrological soil groups B, B/C, C and C/D, which rank soils according to their runoff potential when thoroughly wet, from moderately low in group B to high in group D (Schulze & Schütte, 2023; USDA-NRCS, 2009:7-2). Group C/D lies in the north-east and along the northern edge, group B/C in a belt from the west across the centre, group C in the south, and group B in one small patch in the north-east (Map 5). Natural woodland covers most of A31E, with natural grassland in the south and west. Fallow land, old fields and cultivated land lie mainly along the valley, with small patches of irrigated land, and plantation forest lies in the south. Kromellenboog Dam lies in the north-east (Map 6; DFFE, 2024).



Map 5: Hydrological soil groups of A31E according to the SCS classification as applied in South Africa (Schulze & Schütte, 2023).



Map 6: Land cover of A31E in 2022 (DFFE, 2024).

2.6 Water use and water quality

Of the 3.38 Mm³/a registered in A31E, 69.3% is registered as "scheme" use, 22.2% as river abstraction and 8.4% as borehole use, and 98.9% of the total is for irrigation (DWS, s.a.f). The Department noted an increasing return flow from irrigation in A31D and from the mines around Zeerust, whose nutrient and salt content has limited its re-use for domestic supplies without

costly treatment (DWAF, 2004:57). It also described the return flow from the towns and from irrigation as "being discharged into the Klein Marico River" (DWAF, 2004:86). Water quality was monitored at seven sites in A31E between 1966 and 2023 (DWS, s.a.e).

3. Literature foundation

3.1 Reliability of supply from a small dam

The performance of a water-supply system is judged by more than the average volume that it delivers. Hashimoto et al. (1982:14–17) proposed three criteria for this purpose. Reliability describes how often the system is in a satisfactory state, resiliency describes how quickly it recovers once it has failed, and vulnerability describes how severe a failure is. All three criteria depend on a threshold that separates a satisfactory delivery from an unsatisfactory one. In this study, a failure is a period in which the canal delivers less than the registered scheme volume, so that the criteria answer directly how often, for how long and by how much deliveries fall short. Reliability of the supply is not the same as the resilience of the irrigators, who may plant a smaller area or apply less water when the supply decreases (Lankford et al., 2023).

A small dam can carry a scheme through one dry year, but not through a run of dry years. The relationship between rainfall and runoff can shift during a multi-year drought (Saft et al., 2015), so that a relationship calibrated on wetter years may overestimate the inflow in precisely the years that determine reliability. Evaporation adds to this problem, since the loss increases with the surface area of the water and is expected to increase further as the climate warms, and it is often overlooked in the water accounting of small reservoirs (Aminzadeh et al., 2024). The registered volume is used as the threshold, because it is the volume that the scheme is entitled to receive, but it differs from both the water gauged at the head of the canal and the water used on the fields (Section 6.1).

The only planning assessment found for the dam is the Department’s strategy report for the Marico, which gives a yield without a stated method, does not include the dam in its proposed systems model and does not mention the drought at the dam (DWAF, 2004:50, 83–84). Its finding that the Marico sub-area was “in balance” while the Reserve was not implemented (DWAF, 2004:53, 93) concerns the whole sub-area, and it says nothing about how often the supply of this particular dam fails.

3.2 Groundwater as a backup, and modelling with few data

Groundwater security during drought depends on the water stored in an aquifer, on access to this water through boreholes and on the demand placed on it (Calow et al., 2010:246). A source that is reliable for domestic supply may fail under the larger demand of irrigation. Sustainable groundwater use should be based on capture, which is the decrease in discharge and increase in recharge that pumping brings about, rather than on recharge alone (Seward et al., 2006:473). Where groundwater discharges to rivers, capture is largely a loss of streamflow, and across the United States it supplied about 0.85 of long-term withdrawals, with a mean of 0.61 in the 34 aquifer systems and areas that Konikow and Leake (2014) reviewed. This loss arrives later than the pumping and continues after it stops (Leake & Barlow, 2013), so that a balance of recharge

and use cannot show which discharge an abstraction would reduce. In A31E, access to groundwater appears to be the main limitation, because the irrigated land lies on surface deposits of unknown thickness in the lower mapped yield class (Section 6.1).

Where records are sparse, models are often calibrated over short periods. Ebrahim et al. (2019) calibrated a groundwater model of the hard-rock Hout catchment over five years and estimated the unmetered pumping from satellite data. A model calibrated over a few years may not represent a drought that is longer than its calibration period. For Klein Maricopoort Dam, the monthly reservoir record covers 1936 to 2010 and includes the long drought, and the simplest defensible approach uses this record directly. For groundwater the records are too few to identify a single conceptual model, and several should then be carried forward (Barnett et al., 2012:26), because calibrated models that differ little in their heads can differ greatly in their predictions (Troldborg et al., 2007:843).

3.3 Climate change and the canal supply

The projections used in this study are those assessed by the IPCC in its Sixth Assessment Report. The Atlas of this report gives regional results for the global models of the sixth Coupled Model Intercomparison Project (CMIP6) and for the regional models of CORDEX-Africa (Gutiérrez et al., 2021). A31E lies in the reference region of ESAF (Iturbide et al., 2020), about 1° east of its boundary with western southern Africa. The projections for ESAF are used because the catchment lies inside this region, but its regional mean is weighted towards the wetter east. For 2041–2060, the CMIP6 models under an intermediate emission scenario (SSP2- 4.5) give a median change of about −10% in the rainfall of September to November, and 91% of the models are drier (calculated from Gutiérrez et al., 2021; Iturbide et al., 2020, 2022). The IPCC also projects an increase in agricultural and ecological drought in southern Africa (Ranasinghe et al., 2021:1771).

Projections that were downscaled for South Africa are less uniform. The Green Book projects that the warming may exceed 3 °C over the northern parts of the country by 2021–2050 under high emissions, and it projects more rain over the central interior but drier conditions in the north-east in most of its runs (Engelbrecht, 2019:12, 37). The ensembles agree on strong warming and differ on rainfall. Because perennial drainage responds more than proportionally to rainfall where about 500 mm falls per year (De Wit & Stankiewicz, 2006), even a modest drying could reduce the inflow to the dam by a larger fraction, which is an inference of this study. The wettest tenth of the CMIP6 models gives about +0.5% to +3% more annual rain (calculated from Gutiérrez et al., 2021). The climate cases of this study are drawn from the median and the dry end of the projections.

3.4 Design floods and contaminant pathways

Design floods in South Africa are estimated by three groups of methods (Smithers, 2012:634). Statistical methods analyse observed floods, deterministic methods convert a design rainfall into a flood, and empirical methods are based on the largest floods recorded in a region. Statistical analysis requires a record of observed floods that is “adequate in both length and quality” (Smithers, 2012:635), which the monthly records of A31E do not contain. The Standard Design

Flood (SDF) is a deterministic method that was developed from the flood records of 152 gauging stations in 29 drainage basins (SANRAL, 2013:3-43, 3-44). It exceeded statistical estimates by 60% on average and is regarded as conservative (Smithers, 2012:639), and it is the design flood used in this study.

Contaminants are approached through the source–pathway–receptor model, in which water is at risk only where a source releases a contaminant, a pathway carries it and a receptor receives it (Mbae et al., 2024). The pathway through the ground can be traced by particle tracking. The program MODPATH follows particles through the flows calculated by MODFLOW and derives their times of travel from the porosity of the material (Pollock, 2016:2–3). Particle tracking shows where water goes and roughly how long it takes, but not how a dissolved substance spreads or reacts.

The literature provides the criteria, the reasons why multi-year droughts and evaporation govern a small dam, a projected range of climate change and methods for screening floods and pathways. It does not provide a measure of how often, for how long and by how much the deliveries from Klein Maricopoort Dam fall short, how this would change by about 2050, or whether groundwater in A31E could make up part of a shortfall. This study addresses that gap.

4. Conceptual model

4.1 The system and its two sources of water

The system studied is the land between two dams, together with the upper dam that supplies it (Figure 1). Water enters this system in two ways, which are referred to as the two waters. River flow from the catchments upstream is caught in Klein Maricopoort Dam, and rain falls on A31E itself. The first water feeds the canal and the irrigated land, while the second can reach neither the upper dam nor the canal. Path 1 carries the water from outside A31E through the upper dam, where evaporation removes water but not the salts dissolved in it, to the canal and the river. Path 2 begins with the rain on A31E and runs over the ground to the streams or down to the groundwater. The two paths meet below the canal, on the irrigated land, where canal water drains to the river and the groundwater, and in Kromellenboog Dam, which receives both. The inflow to the upper dam is taken as recorded and serves as the condition at the upstream boundary.

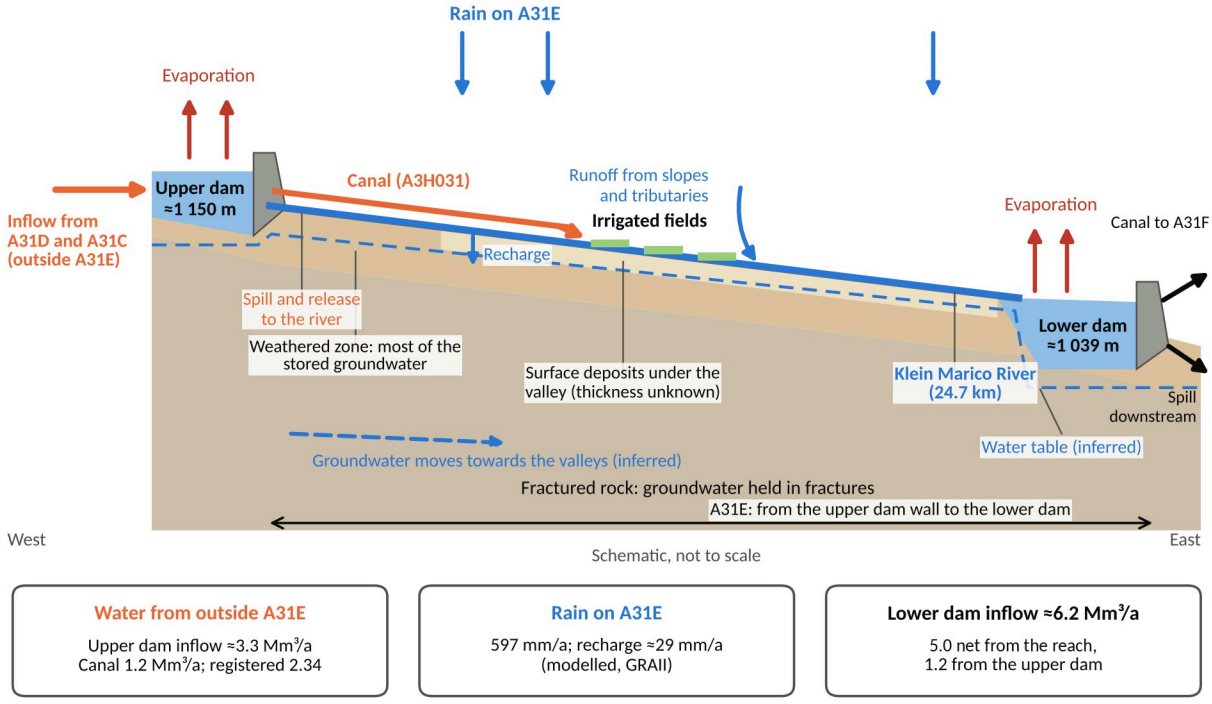


Figure 1: Schematic section of A31E along the Klein Marico River, from the wall of Klein Maricopoort Dam to Kromellenboog Dam, showing the water that enters from outside A31E (orange) and the water from rain on A31E itself (blue). Quantities are means for hydrological years 1987–2009; not to scale.

4.2 The upper dam and the canal

Water leaves the upper dam by four routes, namely evaporation, the canal, the spill and the release to the river, and no supply to Zeerust appears in the record. Beyond the gauge at its head, the canal is not measured. Its losses, the fields that it serves and the water that returns to the river or sinks to the groundwater are unknown, but none of this water can return to the upper dam. In a year of median inflow, a delivery close to the registered volume lowers the storage,

while in a wet year most of the water spills, because the dam is already full. The storage of the dam and the inflow from upstream respond over several years, and these slow parts of the system decide whether the canal can deliver.

4.3 Rain on A31E, groundwater and the lower dam

The runoff of A31E is gathered by the tributaries and ends in Kromellenboog Dam. Most of the rain that does not run off evaporates, and a small part recharges the groundwater, which is stored mainly in the weathered zone. That this groundwater moves from the high ground towards the valleys is an inference from four single water levels (DWS, s.a.c). Groundwater leaves the catchment through boreholes, as discharge to the streams, by uptake of vegetation and possibly as underground outflow (DWS, 2021).

Kromellenboog Dam receives the water of both paths. The net contribution of the reach between the dams is mostly runoff and groundwater discharge of A31E, but it also contains any canal water that returns to the river from the fields, less the water abstracted from the river.

4.4 Contaminant sources, pathways and receptors, and floods

The same conceptual model shows the routes along which dissolved substances would travel (Section 3.4). Two main sources can be identified. The first is the return flow from Zeerust, from irrigation in A31D and from the mines around Zeerust, whose nutrient and salt content has limited its re-use (DWAF, 2004:57). Since Zeerust and A31D lie above the dam (Map 1), this return flow is inferred to enter the river upstream of it. The second is the irrigated land below the dam, where water not used by crops carries salts, and anything applied to the fields, back to the river or down to the groundwater. The rest of the catchment had a population of only 936 in 2011 and contains no large urban source (DWS, 2021:156).

These sources reach the receptors along path 1 and path 2. Below the fields, groundwater can discharge to the river only where the water table stands above the river bed, and where it stands deeper, the water leaves by evaporation and uptake by plants instead. Whether the water table stands above the river bed separates the two model structures of Section 5.2.2.

Four receptors follow from these paths: the users of the canal, who receive path 1 water; the groundwater under the irrigated land, which this study assesses as a backup to the canal and for which no quality record exists; the river between the dams; and Kromellenboog Dam, which receives both paths. The river between the dams is an ecological flow site whose category, C (moderately modified), is to be maintained (DWS, 2021:130; Kleynhans & Louw, 2007:A2-11). The DRASTIC index indicates how easily water from the land surface reaches the groundwater (Aller et al., 1987:13). The index is a relative evaluation tool that does not replace on-site investigation (Aller et al., 1987:11, 20), and it says nothing about the surface deposits in particular.

Floods follow the same two paths. A flood from upstream enters as spill over the wall of the upper dam, the tributaries add floods from A31E itself, and Kromellenboog Dam receives both. The Department identified the spillway of Klein Maricopoort Dam as “a major risk” and the spillway capacity of Kromellenboog Dam as “a major risk factor” (DWAF, 2004:170–171).

5. Methodology

The methodology is described in four steps, namely the data, their preparation, their processing within software and their analysis (Sections 5.1 to 5.4). Each step deals in turn with the water balance of the dam, groundwater, floods, and water quality and climate. All the methods rest on the conceptual model of Section 4, and the water balance uses monthly records.

5.1 Data

5.1.1 Dam water balance

The station records of the Department give one value per month with a quality code (DWS, s.a.b). The reservoir record gives the storage, outflows, evaporation and rain of Klein Maricopoort Dam for every month from 1936 to 2010 (DWA, 2012), and the WR2012 study supplies the rainfall of zone A3B and the patched inflow (Bailey & Pitman, 2016). Evaporation at the dam is measured with a Symons pan (S-pan). The inflow is not gauged but calculated from the dam's own water balance (Bailey & Pitman, 2016:30; DWA, 2012). The only record of runoff generated inside A31E is that of the Wilgeboomspruit (station A3H016), which drains 39 km² and ends in 1992 (DWS, s.a.b). The national register of water use (DWS, s.a.f) records entitlements and not the actual abstraction. The monthly range of each record is listed in Table 1.

Table 1: Stations and records used in this study, with the range of their monthly data and, in brackets, the number of months (Bailey & Pitman, 2016; DWA, 2012; DWS, s.a.b).

Station	What it measures	Monthly data
A3R002	Spill over the wall of Klein Maricopoort Dam	1936/06–2026/06 (1 081)
A3R002	Storage in the dam (reservoir record)	1936/06–2010/11 (894)
A3H031	Volume in the canal	1936/06–2026/05 (1 080)
A3H030	River below the dam (low flows only)	1987/02–2011/05 (292)
A3H042	River below the dam	2012/01–2026/06 (174)
A3E003	Rainfall at the dam	1967/09–2025/02 (690)
A3E003	S-pan evaporation at the dam	1967/10–2025/02 (689)
A3H016	Flow in the Wilgeboomspruit (39 km²)	1965/12–1992/08 (321)
A3R003	Spill over the wall of Kromellenboog Dam	1955/02–2026/07 (858)
A3H033	Volume in the Kromellenboog canal	1955/02–2026/07 (858)
A3H032	River below Kromellenboog Dam	1979/12–2026/07 (560)
Zone A3B	WR2012 rainfall (not a station)	1920/10–2010/09 (1 080)

The Department assigns a quality code to every month. Table 2 gives the share of months in each code for the gauges at Klein Maricopoort Dam, and Table 3 gives it for the Wilgeboomspruit and the gauges at and below Kromellenboog Dam. The spill and canal records at the upper dam have about 74% good continuous data. The S-pan record has 37.0% good continuous data and 43.5% estimates, and the river gauge A3H042 has 5.2% good continuous data.

Table 2: Data quality of the monthly records at Klein Maricopoort Dam, in per cent of the months in each DWS quality code (calculated from DWS, s.a.b). The number of months is in brackets.

Quality code	A3R002 spill (1 081)	A3H031 canal (1 080)	A3H030 river (292)	A3H042 river (174)	A3E003 rainfall (690)	A3E003 S-pan (689)
Good continuous data	73.7	73.8	71.9	5.2	61.6	37.0
Good edited data	6.0	1.1	0.0	21.3	0.0	0.0
Good monthly reading	0.0	0.0	0.0	0.0	21.7	0.2
Audited gauge plate readings	4.4	0.7	0.0	0.0	0.0	0.0
Height derived from flow	0.0	9.6	0.0	0.0	0.0	0.0
Gap filled data	2.4	0.0	0.0	0.0	0.0	0.0
Audited estimate	0.4	0.0	2.1	0.0	0.0	40.1
Minimum value	0.0	0.0	5.1	0.0	0.0	0.0
Above rating	0.0	0.0	7.2	0.0	0.0	0.0
Permanent gap	2.9	3.5	2.7	10.3	0.0	4.4
Data missing	1.2	1.3	0.0	36.2	8.6	3.8
No info for stage/discharge determination	0.1	0.0	0.0	0.0	0.0	0.0
Data unreliable	0.0	0.2	0.0	0.0	0.0	0.6
Negative	0.0	0.0	0.0	0.0	0.0	0.2
Undefined code 8	0.3	0.0	0.0	0.0	0.0	0.0
Previous week’s level used as estimate	0.5	0.0	0.0	0.0	0.0	0.0
Edited and checked – still unaudited	4.8	2.0	0.0	17.2	0.0	0.0
Checked – still unaudited	0.0	4.8	0.0	0.0	7.0	4.9
Checked estimate – still unaudited	0.7	0.6	0.0	0.0	0.0	3.5
Unaudited	0.2	0.0	10.3	0.0	1.2	5.5
Good edited unaudited	2.5	2.4	0.7	9.8	0.0	0.0

Table 3: Data quality of the monthly records of the Wilgeboomspruit and at and below Kromellenboog Dam, in per cent of the months in each DWS quality code (calculated from DWS, s.a.b). The number of months is in brackets.

Quality code	A3H016 Wilgeboomspruit (321)	A3R003 spill, Kromellenboog (858)	A3H033 canal, Kromellenboog (858)	A3H032 river below Kromellenboog (560)
Good continuous data	79.4	66.2	69.5	51.3
Good edited data	0.0	8.6	9.3	15.7
Audited gauge plate readings	0.0	7.8	1.1	0.7
Audited estimate	0.0	0.0	1.4	1.4
Minimum value	1.9	0.0	0.0	0.0
Above rating	0.0	0.0	0.1	0.4
Permanent gap	18.4	1.4	6.5	11.3
Data missing	0.3	0.2	1.3	2.0
No info for stage/discharge determination	0.0	0.4	0.0	0.0
Data unreliable	0.0	0.1	0.0	0.0
Unknown	0.0	0.0	0.1	0.0
Undefined code 8	0.0	4.7	0.0	0.0
Previous week's level used as estimate	0.0	2.0	0.0	0.0
Edited and checked – still unaudited	0.0	5.8	6.9	12.0
Checked – still unaudited	0.0	0.0	0.7	0.4
Checked estimate – still unaudited	0.0	0.1	0.1	0.2
Good edited unaudited	0.0	2.7	3.0	4.8

5.1.2 Groundwater

The borehole records are those of the NGA (DWS, s.a.c). The NGA holds 425 boreholes inside A31E. Each of the 266 boreholes with a water level has a single level, measured between 1910 and 1994, and four of these are recorded as static water levels, three of them during the drought (Table 4). No water-level series or pumping tests exist.

Table 4: The four boreholes inside A31E with a static water level, with the date of the level (DWS, s.a.c), and the mapped classes in which they lie (measured in GIS).

Borehole	Date of level	Depth to water (m)	Rock group	Aquifer type	Mapped yield class	Soil group
2526AD00180	11 May 1994	30.00	Volcanic rocks, diabase and tuff	Intergranular and fractured	0.5–2.0 L/s	B/C
2526CA00602	9 July 1990	9.24	Quartzite and sandstone	Fractured	0.5–2.0 L/s	B/C
2526CA00864	10 July 1990	20.23	Surface deposits	Intergranular and fractured	0.1–0.5 L/s	B/C
2526CB00135	25 November 1974	10.97	Quartzite and sandstone	Fractured	0.5–2.0 L/s	C

The mapped layers are the 1:250 000 geological maps (Michaluk & Keyser, 1991; Walraven, 1981), the 1:500 000 hydrogeological map with its borehole-yield classes (Barnard, 1999), the hydrological soil groups (Schulze & Schütte, 2023) and the national land cover of 2022 (DFFE, 2024), which is classified from satellite images. The Groundwater Resource Assessment II (GRAII) gives the following modelled regional values for A31E (DWS, s.a.a). Transmissivity, which is how easily the full thickness of an aquifer passes water, is 22.81 m²/d. Storativity, which is the water released per m² of aquifer when its level falls by 1 m, is 0.00998 in the weathered zone and 0.00014 in the fractured zone below it. The weathered zone is 25.36 m thick and the fractured zone 182.14 m, a total of 207.5 m. Recharge, which is the part of rainfall that reaches the groundwater, is 4.84% of rainfall, about 29 mm/a. The Reserve determination gives the recharge, use and baseflow of A31E (DWS, 2021). The vulnerability of the aquifer is taken from the national DRASTIC map (DWAF, cited by Esterhuyse, 2017).

5.1.3 Flood

The terrain of the flood model is the 30 m Shuttle Radar Topography Mission (SRTM) elevation model of February 2000 (Farr et al., 2007), and the roughness follows the land-cover groups of the national land cover of 2022 (DFFE, 2024). The design rainfall, the runoff coefficients and the flood formulas are those of the drainage manual of the South African National Roads Agency (SANRAL, 2013).

5.1.4 Water quality and climate

The water-quality records are the analyses of the Department at seven sites in A31E (DWS, s.a.e). The sites, in order along the Klein Marico River, with the period of each record and the number of samples with an electrical conductivity (EC), are: A3H030, the river at the weir below Klein Maricopoort Dam (1989–1993; 56 samples); A3H042Q01, the river below the dam (2012–2023; 81); 188257, the river at Kalkdam, Enzelsberg road (2005–2007; 3); 188038, the river at Nooitgedacht (2004–2005; 3); 188030, the river at Dammenbrug (2004; 1); A3R003Q01, Kromellenboog Dam near the wall (1966–2023; 248); and A3H016Q01, the Wilgeboomspruit (1971–1991; 54). Path 1 is sampled below Klein Maricopoort Dam and at the three sites along the river between the dams, path 2 in the Wilgeboomspruit, and both paths in Kromellenboog Dam. The groundwater of A31E is represented by the median electrical conductivity of seven samples given in the Reserve determination (DWS, 2021:163). EC, which measures the dissolved salts, chloride, sodium, the sodium adsorption ratio (SAR), inorganic nitrogen and phosphate-P are screened against guideline values, and sulphate is reported as a median. The guideline values are the upper limits of the target range for irrigation, where the SAR of 2.0 is the value for the crop yield, and the boundaries of the trophic classes for aquatic ecosystems (DWAF, 1996a:134; DWAF, 1996b).

The climate cases use the CMIP6 projections of the IPCC Atlas for ESAF for 2041–2060 under SSP2-4.5 (Gutiérrez et al., 2021; Iturbide et al., 2020, 2022).

5.2 Data preparation

5.2.1 Dam water balance

Rainfall and flow are counted by hydrological year, which runs from October to September and is named after the calendar year in which it begins. Months that were partly recorded are treated as gaps, and a yearly total is counted only for a year in which every month has a value. Three periods are doubtful although their codes are good. These are the calculated inflow of 1975 to 1980, which totals 296 Mm³ in the patched WR2012 flow at the dam against 186 Mm³ at Kromellenboog Dam downstream, the spill record of 1977 to 1979, and the canal record of 2018 to 2021, which repeats one flow rate (DWA, 2012; DWS, s.a.b).

The water-surface area follows from the relation between the gauge reading and the contents of the reservoir since the survey of 1982, which gives the capacity of 7.07 Mm³ at full supply on which the reservoir record is based. Evaporation is calculated from the twelve mean monthly S-pan depths, because 43.5% of the pan months are estimates, with pan factors of 0.82 to 0.89 that were back-calculated from the Department’s evaporation volumes. The release to the river is taken as the mean gauged release in each band of storage (Figure 2, upper panel).

The upper panel uses the 259 months without spill and shows the mean of each storage band. The lower panel gives the share for the fit period (hydrological years 1966–1974 and 1981–1986), for the test period (1987–2009) and for both together, which is the rule used.

The canal is asked to deliver the registered volume of 2.34 Mm³/a, which it carried when the dam was full (DWA, 2012; DWS, s.a.b). This volume is divided over the year according to the pattern of Figure 3, which is taken from the canal record A3H031 for the hydrological years 1997–2014 without 2010. About a third of the volume falls from

The inflow of each month is the remainder of the dam's own recorded balance. This remainder is available for 843 months. A further 27 months were rebuilt from the station records and 18 from the WR2012 patched flow, which gives the 888 months from October 1936 to September 2010. This is inflow series 1, and all results are for series 1 unless stated. Before 1987 the release entry of the dam record is a fixed value, and a second inflow series (series 2) that adjusts it was prepared for this period and run as a check. The inflow to Kromellenboog Dam was calculated in the same way, as the remainder of its recorded balance (A3R003, A3H033), and the net contribution of the river between the dams is this inflow less the spill and release of Klein Maricopoort Dam.

No operating rule of the scheme could be obtained. The share of the registered volume that the canal receives when the dam is low was derived from the records, as Coerver et al. (2018) do for reservoirs whose rules are unknown. The rule gives this share for seven zones of storage, calculated as the canal volume delivered divided by the volume asked for in all months whose starting storage lay in that zone. The rule was fitted on hydrological years 1966–1974 and 1981–1986 and tested on 1987–2009, the period in which every outflow is measured (Figure 2, lower panel). The seasonal pattern of the canal volume in the forward test is that of Figure 3, and in the reverse test it is that of the months with the dam at 85% or more of its capacity in 1966–1974 and 1981–1986.

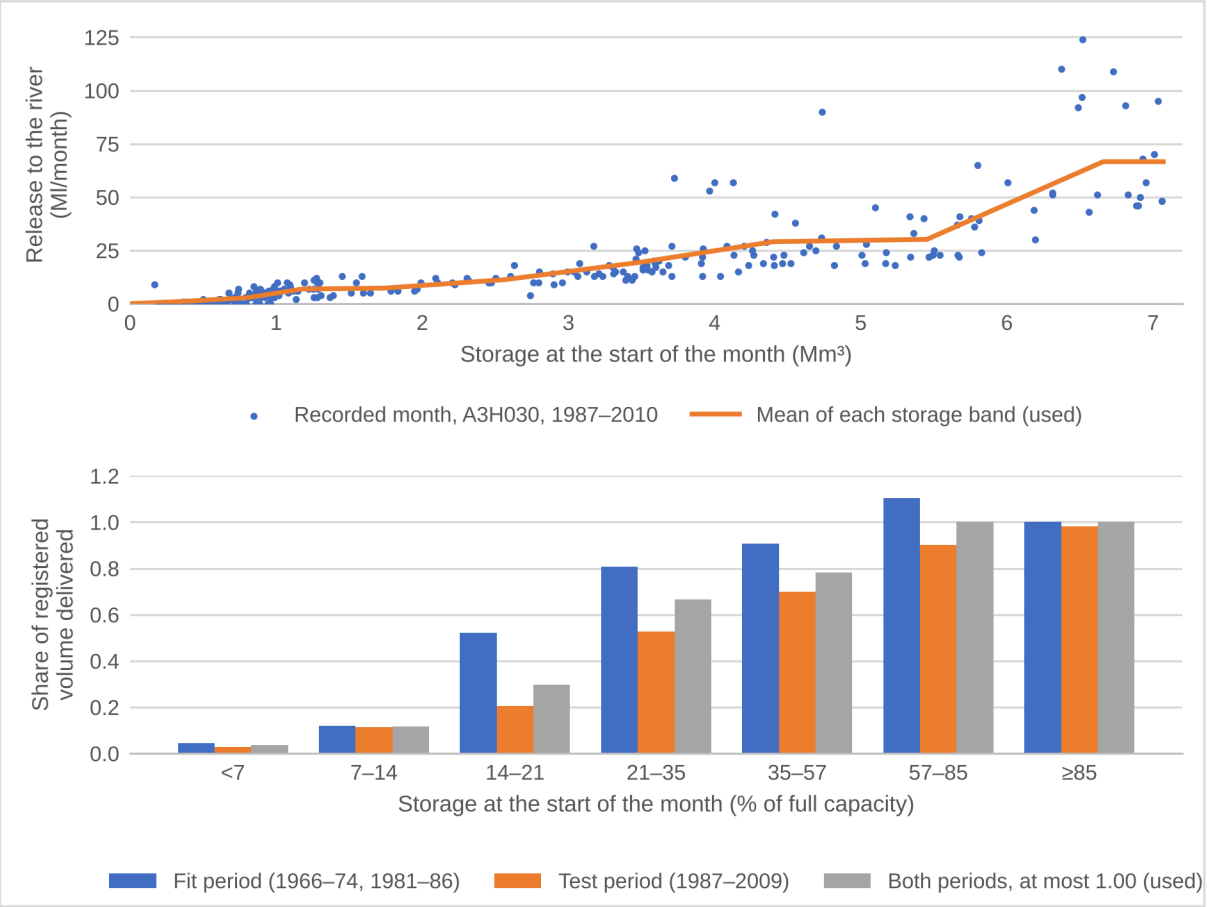


Figure 2: Upper panel: monthly release to the river below the dam (DWS station A3H030) against the storage at the start of the month, 1987–2010. Lower panel: share of the registered volume that the canal received in each storage zone.

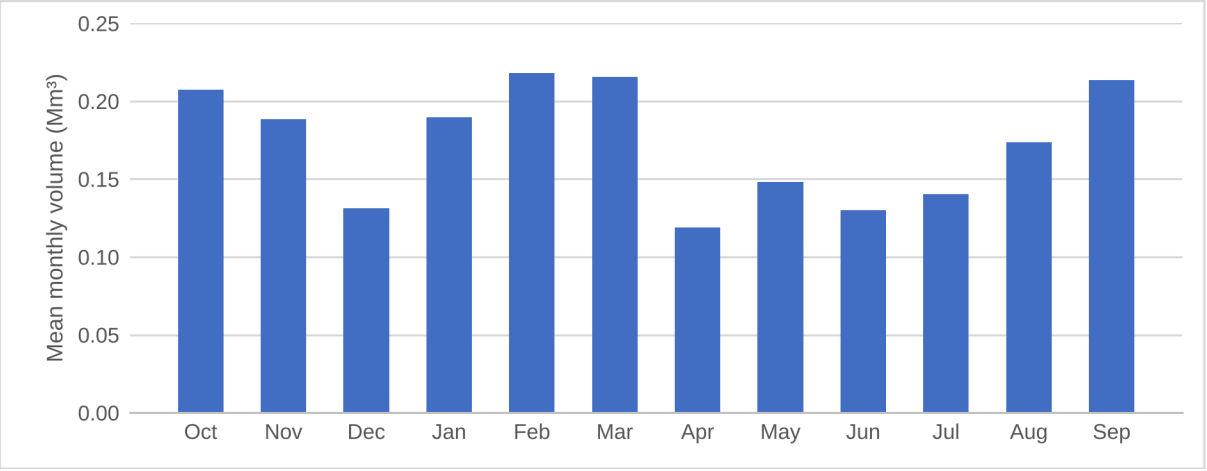


Figure 3: Average monthly volume in the canal from Klein Maricopoort Dam (DWS station A3H031), hydrological years 1997–2014 without 2010.

5.2.2 Groundwater

A 20 m digital elevation model (DEM) was made in ArcGIS Pro from spot heights and contours. It gives the top of the groundwater model, the ground levels of the boreholes and the levels of the drains.

The model removes water from every cell through a ground drain, which represents seepage, springs, evaporation and uptake by plants. Two depths of the ground drain are used, and they define the two structures of the model: structure S, with a shallow ground drain, and structure D, with a deep one. In structure S the ground drain lies 1 m below the land surface. In structure D it lies 15.27 m below the surface, which is the median depth to water of 112 archive levels in A31E and its neighbouring catchments (DWS, s.a.c). These 112 levels include the four calibration levels but none of the 262 levels that are later compared with the model. The depth of 15.27 m is also close to the mean maximum rooting depth of tropical grassland and savanna, 15.0 m (Canadell et al., 1996).

The calibration targets are the four static water levels, and the 262 other archive levels are kept for comparison afterwards. The recharge is the regional rate of 29.2 mm/a. The benchmark for the fit is a water table 15.27 m below the ground everywhere.

The abstraction is placed in GIS in three ways. Placement P1 spreads it over the 45 model cells under the land mapped as irrigated, P2 over the 474 cells of surface deposits within 1 km of that land, and P3 over the 410 bedrock cells within the same distance. Because the effective porosity of A31E is not known, a porosity of 0.10 is used for the particle times, which lies within the range published for fine sand, silt and clay, the surface deposits in which 41 of the 45 cells under the irrigated land lie (Woessner & Poeter, 2020).

5.2.3 Flood

Flood extent and depth were modelled for the reach of the Klein Marico River below the Doringrivier, because the irrigated land lies along it. The reach begins at point F2 and runs 15.7 km downstream (Map 2). The catchment at F2 covers 1 357 km², including the catchment of Klein Maricopoort Dam (F1) and that of the Doringrivier (F3), and its watercourse length and slope were taken from GIS.

The peak of the SDF was calculated in the steps of the drainage manual (SANRAL, 2013:3-44 to 3-47, 3B-2, 3B-3), with the inputs in Table 5. The catchment at F2 lies in SDF basin 1. Its area, the length of its longest watercourse and the slope between 10% and 85% of that length were taken from the GIS. The design rainfall for the time of concentration was interpolated linearly between the modified Hershfield value for 6 h and the daily value of the basin station, Struan (SAWS 546 204), and reduced by the area reduction factor. The runoff coefficient for each return period follows from the coefficients of the basin for the 1:2-year and 1:100-year floods through the return-period factors. Each peak was given the triangular hydrograph that the manual prescribes for the SDF, which rises over one time of concentration and falls over two (SANRAL, 2013:3-45, 3-47).

The SRTM model is used for the flood model alone, because the 20 m DEM smoothed the valleys. Manning’s n was set for each land-cover group within published ranges (USACE HEC, s.a.b).

Table 5: Inputs of the SDF at F2.

Step	Quantity	Value
1	SDF basin; station	1; Struan (SAWS 546 204)
2	Catchment area, A	1 357.26 km²
3	Longest watercourse, L	94.12 km
4	Average slope between 10% and 85% of L, S	0.00536 m/m
5	Time of concentration, Tc = (0.87 L² / (1000 S))^0.385, L in km	16.44 h
6	Point rainfall for Tc, 1:50-year and 1:100-year (modified Hershfield at 6 h: 122.1 and 140.7 mm; daily value of Struan: 150 and 177 mm)	138.3 and 161.8 mm
7	Area reduction factor	84.4%
8	Runoff coefficient C2 and C100; C50	10% and 40%; 36.4%
9	Peak, Q = C I A / 3.6, with I = point rainfall × ARF / Tc	Table 14

5.2.4 Water quality and climate

Surface samples were used for the water-quality screening.

The climate cases are expressed as factors that multiply the value of the Reference run, which is the water balance and the groundwater model with inflow series 1, the evaporation and demand of the records and the regional recharge. They are sensitivity steps for about 2050 and not forecasts. Case M represents the median of the projections for ESAF for 2041–2060 under intermediate emissions (SSP2-4.5). Case D represents their dry end, with the 10th percentile of the rainfall and the median warming, because the projected warming is narrow in range while the rainfall is uncertain even in sign (Section 3.3). The factors are given in Table 6.

The cases follow the projections of the models under SSP2-4.5 for ESAF for 2041–2060, set against hydrological years 1937–2009.

Table 6: The climate cases for about 2050 (calculated from Gutiérrez et al., 2021; Iturbide et al., 2020). Each factor multiplies the value of the Reference run.

Case	Warming	Annual rain	Inflow	Evaporation and demand	Recharge
Reference	–	–	1.00	1.00	1.00
M (median)	+1.9 °C	−3.1%	0.93	1.064	0.96
D (dry end)	+1.9 °C	−8.5%	0.80	1.064	0.89

The inflow factors follow from the rainfall elasticity of the inflow, which was estimated at 2.0 to 2.5 with the non-parametric estimator of Sankarasubramanian et al. (2001:1775). Because the inflow stayed below its median from 1982 to 1994 although the rainfall was above its mean from 1987 to 1990, this elasticity is an approximate scaling rather than a property of the catchments above the dam. The recharge factors follow from the finding that recharge across Africa increases with rainfall to a power of about 1.34 to 1.39 (MacDonald et al., 2021). The factor on evaporation and demand corresponds to an increase in potential evapotranspiration of about 3.4% per °C, which lies in the upper half of the “about 1.5–4%” per °C found by Scheff and Frierson (2014:1554), and Cullis and Phillips (2019:25) likewise scale urban water demand by the change in potential evaporation.

5.3 Data processing within software

5.3.1 Dam water balance

The monthly water balance of the dam and its canal was built in Microsoft Excel from the station and reservoir records. It is a mass balance of one reservoir with a restriction rule, which is the form that Adeloye et al. (2016) use to calculate reliability, resilience and vulnerability for a single dam. A simpler storage–yield calculation would leave out evaporation, while a river-system model such as WEAP (Dlamini et al., 2023) would add components for which no records exist.

In each month, the storage at the end equals the storage at the start, plus the inflow and the rain on the water surface, minus evaporation, the release to the river, the canal delivery and the spill. The balance runs over the 888 months from October 1936 to September 2010 and applies the present capacity of 7.07 Mm³ to every year. It describes the present dam and registered volume under the inflows of the past and is not a reconstruction of history. The canal receives no water once the storage reaches 0.166 Mm³, which is the lowest storage since the survey of 1982.

5.3.2 Groundwater

The shares of rock group, aquifer type, yield class and soil group, and all elevations, were measured in ArcGIS Pro by overlaying the mapped layers on the boundary of A31E and on the land mapped as irrigated.

The groundwater model was built in GMS 10.9.6 and run with MODFLOW-NWT (Niswonger et al., 2011). It consists of one layer, 207.5 m thick, with its top at the 20 m DEM, and 15 142 active cells of about 200 m by 200 m (145 rows and 222 columns). The recharge of 29.2 mm/a (17.53 Mm³/a) is applied to every cell, and the outer edge is closed. Water leaves the model only through drains, which remove water whenever the water level stands above them (Harbaugh, 2005). River drains follow the four named rivers at the level of the lowest land beside the channel, and stream drains follow all other mapped streams at the land surface, both with a conductance of 10 m²/d per metre of channel. The ground drain of every cell has a conductance of 0.05 m²/d per m².

The hydraulic conductivity has two zones, surface deposits (3 479 cells) and bedrock (11 663 cells). It was estimated for each structure with PEST (Doherty & Hunt, 2010) within the limits 0.03–3.0 m/d for the deposits and 0.0005–0.05 m/d for the bedrock, using the four static water levels with an interval of 10 m each. The recharge was held at its regional value, because recharge and conductivity cannot be separated in a calibration on heads alone (Knowling & Werner, 2016). The calibrated model without wells is the base case of each structure.

A31E has no water-level series for a transient model, so the model is run at steady state, in which the water entering the aquifer equals the water leaving it, and the cases are compared with one another. The residual of a borehole is its measured water level minus the simulated one, and the fit is expressed as the root-mean-square error of the residuals.

Particle tracking used MODPATH 7 (Pollock, 2016) in the base case of each structure. Nine particles were released at the water table in each of the 45 cells under the irrigated land, 405 in all, and tracked forward. The effective porosity was entered as 1.0 in every cell, and the travel

times are scaled to a porosity of 0.10 afterwards. Each particle was classed by the drains of the cell in which it ended.

5.3.3 Flood

The floods were routed with HEC-RAS 7.0.1 in an unsteady two-dimensional model (USACE HEC, s.a.a), in which the water can spread in any direction over the valley floor as the flood passes. The model area is 24.3 km², with a mesh of 30 m cells refined to 15 m along the river. Manning's n, which describes how strongly the land surface resists the flow, was applied by land-cover group. The upstream boundary at F2 is the inflow hydrograph, with an energy slope from the terrain, and the downstream boundary is normal depth. The default diffusion-wave equations were used. The hydrographs have values every 15 minutes, with a minimum of 5 m³/s.

5.3.4 Water quality and climate

The factors of Table 6 were applied to the inflow, evaporation, demand and recharge of the water balance and the groundwater model described above.

5.4 Analysis

5.4.1 Dam water balance

To pass the test, the simulated monthly storage and yearly canal volume had to meet the criteria of Moriasi et al. (2007:891) for a satisfactory simulation, which were set for streamflow and are applied here by analogy. These criteria are a Nash–Sutcliffe efficiency (NSE) above 0.50, a ratio of the root-mean-square error to the standard deviation of the observations (RSR) of 0.70 or less, and a percent bias (PBIAS) within ±25%. The inflow itself cannot be tested, because it is calculated from the storage record. For the same reason, the fit of the monthly storage mainly tests the restriction rule and the outflows, while the yearly canal volume is the more independent test. The test is independent only for the restriction rule and the demand, because the area curve, the pan factors and the release table were derived from the test period itself, and passing a split test is in any case "a necessary, rather than a sufficient, condition" for a model to be adequate (Klemeš, 1986:13). After the test, the rule was refitted on both periods together, so that the rule used in all further runs also contains the test period. Its shares rise from 0.036 in the lowest zone to 1.00 above 57% of capacity. The second, adjusted inflow series was run as a check. The rule was also fitted on each period alone, to show how far the reliability depends on the period used. In the reverse test the rule was fitted on 1987–2009 and tested on 1966–1974 and 1981–1986, with the storage at the start of each period taken from the record, and the same three criteria were applied to the 178 months and the 14 years with a counted canal volume.

The supply is judged by the criteria of Hashimoto et al. (1982), calculated with the formulas of Adeloye et al. (2016). A month fails when the delivery is below 99% of the registered volume of that month. Reliability by time is the share of months without failure, reliability by volume the share of the registered volume delivered, and yearly reliability the share of years with the full volume. Resilience is the number of shortfall events, which are unbroken runs of failure months, divided by the number of failure months, and vulnerability is the mean depth of a shortfall. The shortfall of a month is the registered volume less the simulated delivery at the head of the canal,

and it is referred to as the shortfall against the registered volume throughout. Every criterion is a frequency counted in one record of 74 years and not a probability, since synthetic series (Kjeldsen & Rosbjerg, 2004) would require a stochastic model of the inflow. A curve of reliability against demand was run for demands of 0.5 to 3.0 Mm³/a. The historic firm yield is the largest demand that the dam could have met in every month of the record without restriction.

The records cannot distinguish a month in which the canal was restricted from a month in which less water was wanted, and every run is made twice. With the restriction rule, the canal receives the share given by the rule, while without restriction it receives the full volume for as long as water remains in the dam. The unrestricted run gives the upper bound of reliability, because every restricted month counts as a failure and much of the water held back in the dam is lost to evaporation. The run with the rule gives the lower bound for the scheme as it was operated on average over the record, but it is not the lowest possible value, since a rule that restricts more would lower the reliability further (Section 6.3). For resilience and vulnerability the order is reversed, because a restricted canal fails more often but less deeply, and the two bounds are always reported together. The sensitivity of the results was tested by changing one input at a time (Pianosi et al., 2016:216), namely the inflow series, evaporation, the capacity, the release to the river, the rule and the seasonal pattern of demand.

5.4.2 Groundwater

The shortfall was first compared with the groundwater balance of the Reserve determination (DWS, 2021). Such a balance treats recharge as water that is free to be taken, while pumped water is balanced mostly by capture (Section 3.2), and only a flow model can show which discharge is reduced by pumping. With four static water levels and a steady-state calibration, the model belongs to Class 1, the lowest class of confidence in the Australian groundwater modelling guidelines, whose uses include developing coarse relationships between abstraction and impact (Barnett et al., 2012:20–21). Both structures are needed because four static water levels cannot identify a single conceptual model (Section 3.2). The fit was compared with the benchmark, and the 262 other archive levels were compared with both structures afterwards. Neither comparison is a validation (SKM, 2013:18).

Abstraction was run at placement P1 at 0.25, 0.5, 1.0 and 2.34 Mm³/a in structure S and at 0.25, 0.5, 1.0, 2.0 and 2.34 Mm³/a in structure D. Placement P2 was run at 0.5 Mm³/a in both structures and at 2.34 Mm³/a in structure S, and placement P3 at 0.5 Mm³/a in structure S, to test whether spreading the abstraction changes the result. The recharge was also multiplied by the factors of cases M and D, 0.96 and 0.89, both without wells and with 0.5 Mm³/a at P1.

Three results are read from each case and compared with the base case at the same recharge. These are the rate actually pumped, because MODFLOW-NWT reduces the rate of a well when its cell nearly dries (Niswonger et al., 2011), the drawdown in the pumped cells, and the change in the outflow of each group of drains as a share of the rate pumped. The drawdown is the fall of the water table caused by the abstraction, and the lowering is the difference in the water table between two runs with the same abstraction, one at the base recharge and one at the reduced recharge. Both are means of 200 m model cells.

A steady-state drawdown is the state reached if pumping continued indefinitely at the mean recharge. It exceeds the drawdown of a single season of pumping at the same recharge, and during a long drought, when the recharge falls below its mean, it would be larger. The loss of streamflow would in reality arrive later and continue after pumping stops (Section 3.2). Two further comparisons show what the four water levels cannot decide. In one, the recharge, the conductivities and the drain conductances are all halved or all doubled, which gives the same water levels and so fits the four levels equally well. In the other, structure D is run with the conductivities of structure S. The mean shortfall and the rate needed in the longest shortfall event were expressed as shares of the recharge of the Reserve determination and compared with the rates that the model could deliver.

The particle times are those of the moving water alone, without dispersion, retention or decay. They scale in proportion to the effective porosity, which was not measured, and a porosity of 0.01 would make them ten times shorter than at 0.10.

5.4.3 Flood

The design flood was estimated with the SDF method of the drainage manual (SANRAL, 2013). Two plans were run, the 1:50-year and the 1:100-year flood. Two simplifications act in opposite directions, since ignoring the attenuation in Klein Maricopoort Dam makes the flood larger, while leaving out the inflow of the Waterkloofspruit makes the flood below its junction smaller. A cell of the mesh was counted as flooded where its maximum depth was at least 0.01 m. The peak reduction is the fall of the peak between the inflow at F2 and the outflow at the downstream boundary, and the delay is the time between the two peaks. The irrigated land flooded is the area of land mapped as irrigated (DFFE, 2024) that lies in the flooded cells, calculated in a GIS.

5.4.4 Water quality and climate

Each water-quality quantity was compared with its guideline value, and the share of samples above the guideline was counted for each site. A value above a guideline shows that the water could affect its use or its ecosystem, but it does not show that the water is contaminated. The guideline defines the trophic classes for the average summer concentration, so the share of single samples above a boundary is a screening indication and not a trophic classification (DWAF, 1996b:84, 97).

In the climate cases, the dam was run with the inflow multiplied by 0.93 and 0.80 and evaporation and demand by 1.064. The canal is asked for the registered volume raised by the same factor, 2.49 Mm³/a, and the reliability is measured against this volume. The groundwater model was run with the recharge multiplied by 0.96 and 0.89 (Section 5.4.2). The simulated storage of the cases was compared by the share of months in which it stood at or above each level.

6. Results

6.1 The irrigated land and the records

A31E falls from 1 567 m in the south to 1 038 m at Kromellenboog Dam (Map 2). Between the two dams, the river falls about 104 m over a distance of 24.7 km. The Doringrivier joins the Klein Marico River 4.2 km below the dam wall and the Waterkloofspruit 13.7 km below it. The water surface of the upper dam stands at about 1 150 m, while all the land mapped as irrigated below it lies lower, between 1 093 and 1 140 m, so that a canal leaving the dam could reach this land by gravity.

The rock types of Map 3 are combined into four groups: quartzite and sandstone cover 31.8% of A31E, the andesitic lava of the Hekpoort Formation and diabase 24.4%, mudrock, shale and hornfels 20.9%, and surface deposits 22.5%. On the hydrogeological map, 59.8% of A31E is classified as an intergranular and fractured aquifer and 40.2% as a fractured aquifer. Of the land mapped as irrigated, 91.6% lies on surface deposits. The thickness and composition of these deposits are not known. The mapped median borehole yield is 0.5–2.0 L/s over 75.3% of A31E and 0.1–0.5 L/s over the remaining area, which includes three-quarters of the irrigated land. The registered scheme volume is equal to about 74 L/s pumped continuously, which would require about 150 to 740 boreholes at yields of 0.5 to 0.1 L/s. According to the modelled regional values of the GRAII (Section 5.1.2), the weathered zone holds about 91% of the water that would drain if both zones were dewatered (storativity multiplied by thickness).

Each of the 266 boreholes with a water level has a single level. The 262 levels that are not recorded as static have a median depth of 19.4 m, and the median strike yield of 109 boreholes is 0.3 L/s.

The soils of A31E belong to the hydrological soil groups B/C (38.7%), C (31.5%), C/D (28.7%) and B (1.2%). 83.2% of the land mapped as irrigated lies on group B/C. The Wilgeboomspruit yielded about 53 mm/a, or 7.6% of the rainfall, in the six complete years of 1966–1980 (DWS, s.a.b). Natural woodland and grassland cover 87% of A31E and cultivated land 10.2%, more than half of which is fallow. Land mapped as irrigated covers 164.9 ha in 18 patches in the strip below the dam, of which 147 ha are under centre pivots (Map 6). Built-up land covers 0.3%.

Half of the scheme positions of the register lie on higher ground above the level of the dam, which a canal cannot reach by gravity. Three different quantities describe irrigation from the dam. The register lists 442 ha and 2.34 Mm³/a, the land cover maps 165 ha of irrigated land, of which 147 ha is under centre pivots, and the canal carried 1.89 to 2.16 Mm³/a in 2022 and 2024 (DWS, s.a.b).

The yearly rainfall at the dam varied between 314 and 900 mm in 1967–2008 (Figure 4). Every year from 1981 to 1986 fell below the long-term mean, which is the longest such run since 1920 (Bailey & Pitman, 2016; DWS, s.a.b). At the dam, 93% of the rain falls between October and

April, and the S-pan evaporation averages 2 005 mm/a and exceeds the rainfall in every month (Figure 5; DWS, s.a.b). The reservoir loses a mean of 1.07 Mm³/a to evaporation over the record of 1936–2010, and about 2.0 Mm³/a when the dam is nearly full (DWA, 2012).

Only complete hydrological years are shown for the station. Hydrological year 1975 is left out, because its total includes an implausible 670 mm in August 1976.

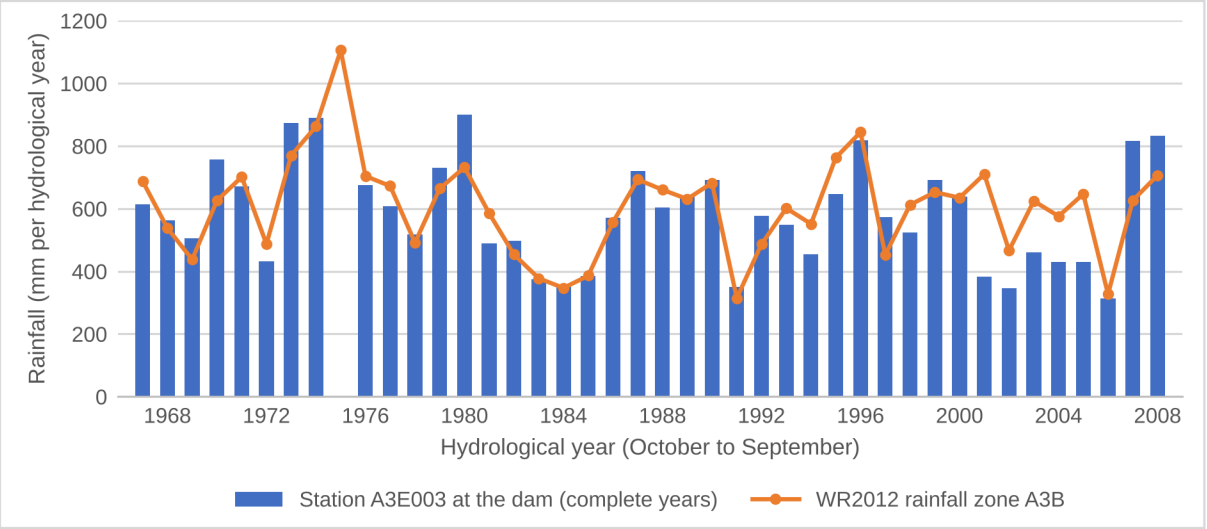


Figure 4: Yearly rainfall at Klein Maricopoort Dam (DWS station A3E003) and in WR2012 rainfall zone A3B, hydrological years 1967–2008.

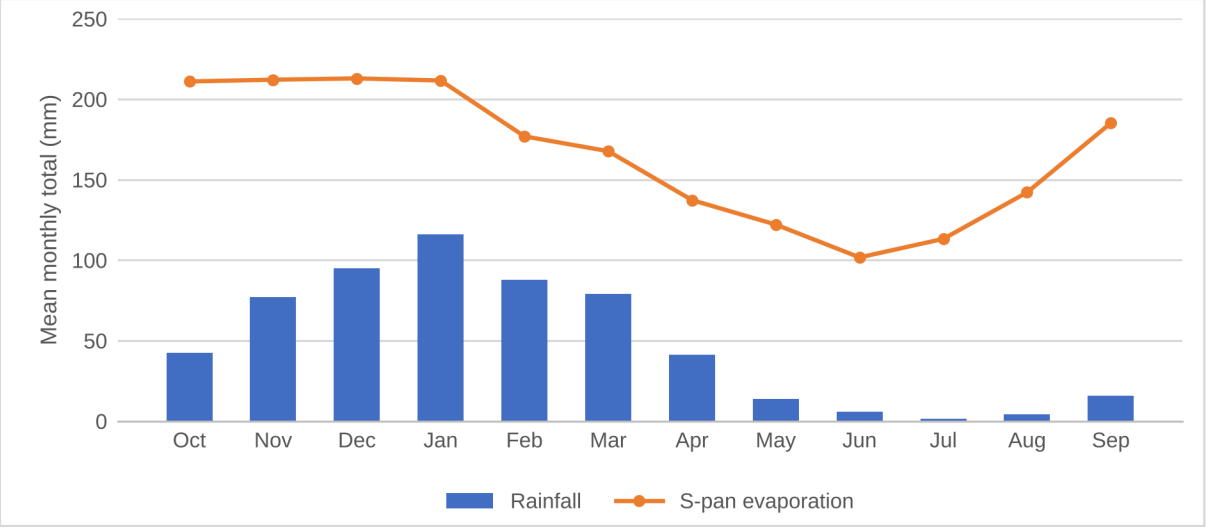


Figure 5: Average monthly rainfall and S-pan evaporation at Klein Maricopoort Dam (DWS station A3E003), hydrological years 1967–2014.

6.2 The canal record and the water of A31E

The inflow to Klein Maricopoort Dam is not gauged, so it was calculated from the dam's own record (series 1). Its median over the hydrological years 1936–2009 is 3.71 Mm³/a (Figure 6). The inflow stayed below the median from 1982 to 1994, and from 1985 to 1994 it averaged 0.84 Mm³/a, about a third of the registered volume before any evaporation. The rainfall was well

below the long-term mean in 1982 to 1985 and in 1991, and above it in 1987 to 1990. The dam fell from nearly full in December 1982 to 22% of its capacity in March 1984 and stayed low until the rains of 1995 and 1996 (DWA, 2012).

The inflow in the lower panel is calculated from the dam's own record (series 1). The lines show the long-term mean rainfall (1920–2009) and the median inflow (1936–2009) (Bailey & Pitman, 2016; DWA, 2012).

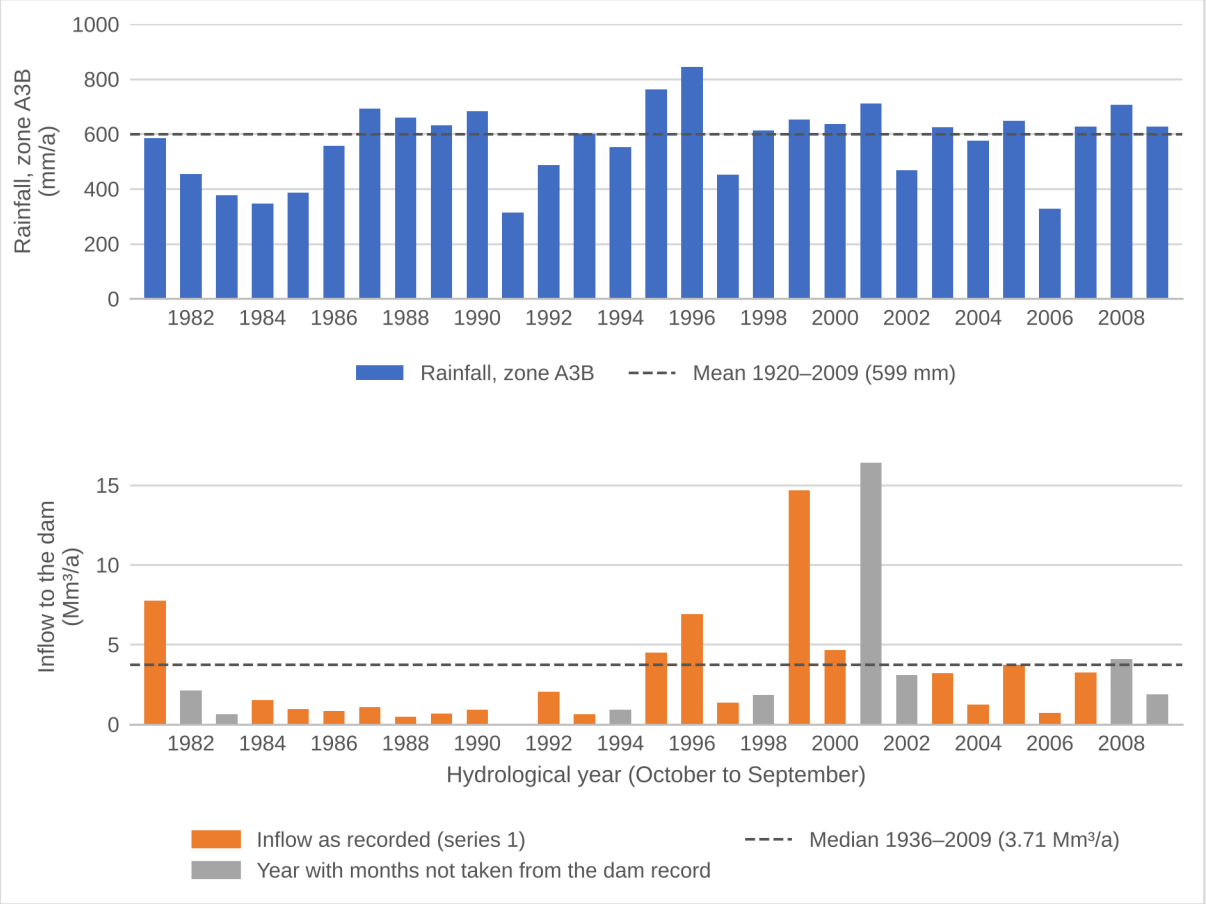


Figure 6: Yearly rainfall in WR2012 rainfall zone A3B (upper panel) and yearly inflow to Klein Maricopoort Dam (series 1, lower panel), hydrological years 1981–2009. Grey: years with months not taken from the dam record.

The water balance shows where the water went (Table 7). Evaporation took a third of the water that entered the dam in 1987–2009 and about half in 1987–1995, when the canal carried only 0.30 Mm³/a. The capacity of 7.07 Mm³ equals about three years of the registered volume, and about two years once the evaporation, net of the rain on the water surface, and the release to the river are included.

The inflow is the remainder that closes the balance of the other terms. The series 1 inflow, which uses the release entry of the dam record, averages 3.39 Mm³/a for 1987–2009. The canal is the irrigation entry of the dam record; gauge A3H031 gives 1.23 Mm³/a for 1987–2009. The columns start in 1987, the first hydrological year in which every outflow is gauged, and the second ends in 1995, the year in which the dam recovered. Rounding leaves the columns up to 0.01 apart.

Table 7: Water balance of Klein Maricopoort Dam from the records, in Mm³/a (DWA, 2012; DWS, s.a.b).

Term	1987–2009	1987–1995 (drought and recovery)	1996–2009
Inflow from the river (remainder)	3.32	1.23	4.67
Rain on the water surface	0.42	0.24	0.53
Evaporation	1.23	0.74	1.55
Canal	1.20	0.30	1.78
Spill	0.87	0.00	1.43
Release to the river	0.33	0.07	0.49
Losses	0.01	0.01	0.01
Change in storage	+0.11	+0.35	−0.05

The canal record shows how often the canal fell short. In 24 of the 38 years from 1966 to 2009 that have a complete record (63%), the canal carried less than the registered volume, by 58% of that volume on average (Figure 7). The years 1984 to 1996 form one run of thirteen shortfall years (DWS, s.a.b).

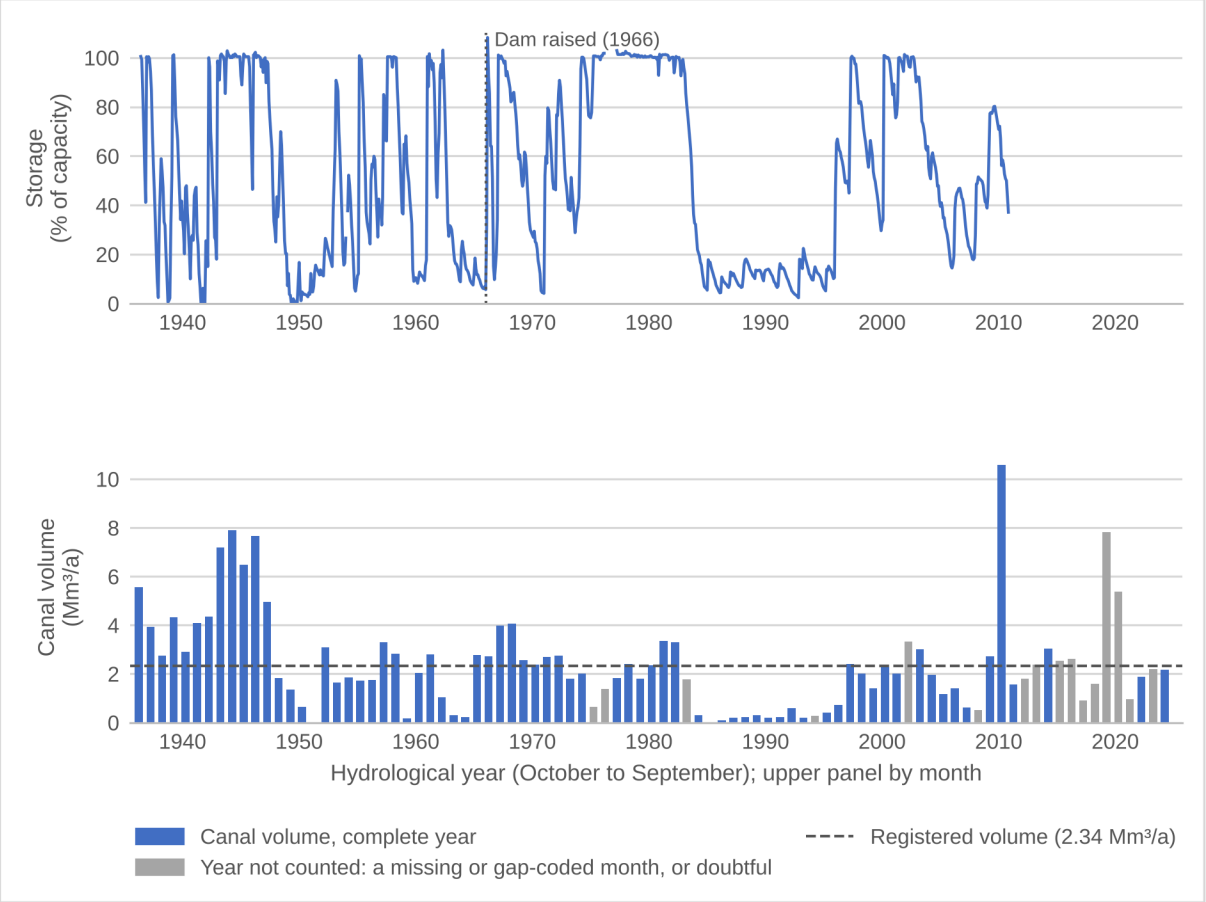


Figure 7: Upper panel: monthly storage as a percentage of the capacity at the time, June 1936 to November 2010 (DWA, 2012). Lower panel: yearly canal volume, hydrological years 1936–2024 (DWS station A3H031), against the registered 2.34 Mm³/a. Grey: years not counted.

To see where the water came from during the drought, the inflow to Kromellenboog Dam was compared with what Klein Maricopoort Dam passed on (Table 8). In 1987–1995 the upper dam passed on almost nothing, and most of the water that reached Kromellenboog Dam came from the river between the dams. Over 1987–2009, 81% of the water that reached Kromellenboog

Dam did not come from the upper dam (DWA, 2012; DWS, s.a.b). The Wilgeboomspruit, the one gauged stream inside A31E, was dry in 42 of the 50 months from February 1984 to March 1988 and flowed again from April 1988 (Figure 8).

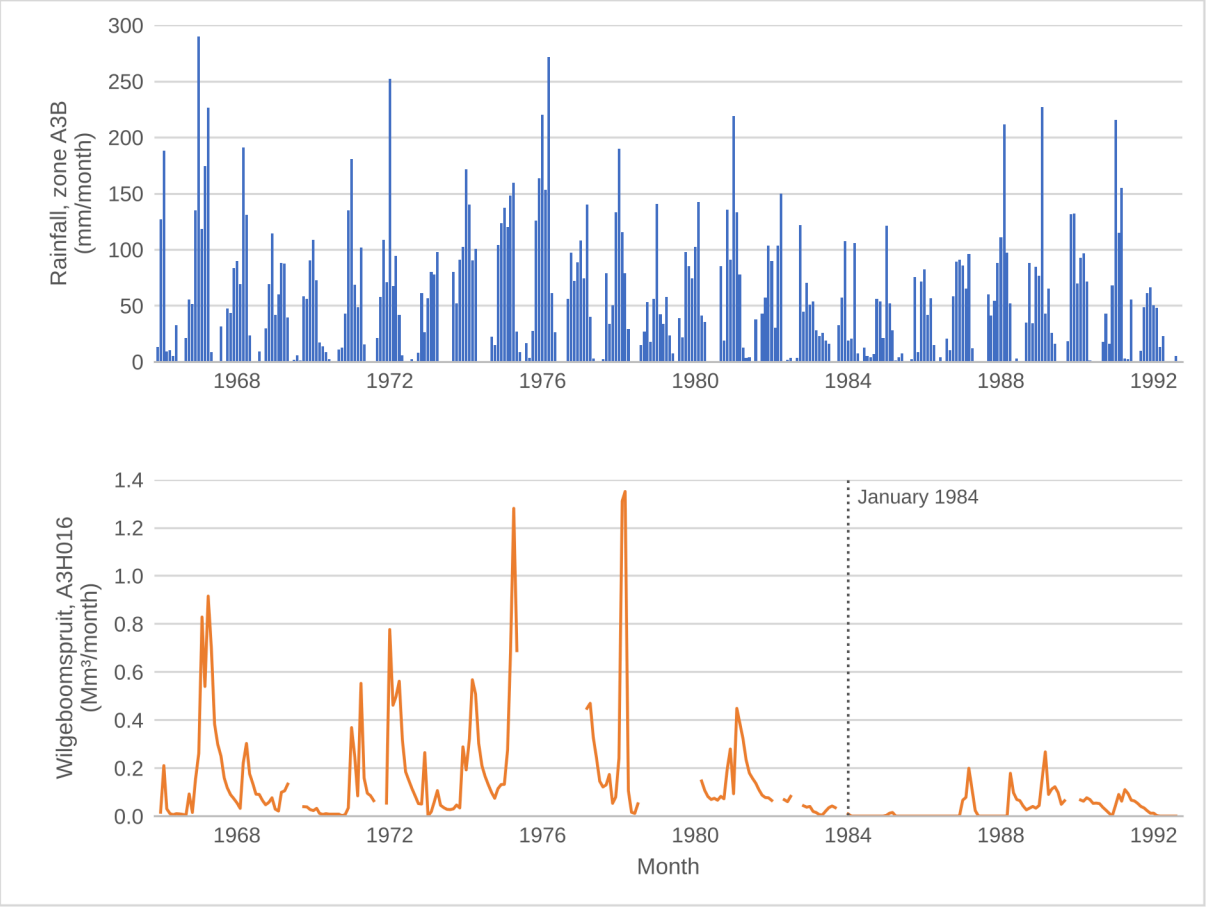


Figure 8: Monthly rainfall in WR2012 rainfall zone A3B (upper panel) and monthly flow of the Wilgeboomspruit at DWS station A3H016 (lower panel), December 1965 to August 1992.

The net contribution is the difference between two remainders of dam records. It is net of river abstraction and includes return flow of canal water, which the records cannot separate. Months with a flagged outflow are included, and the years 1992 and 1993 have no yearly value, so the means use the storage at the ends of the period. Rounding leaves the columns up to 0.01 apart.

Table 8: The inflow to Kromellenboog Dam and the water of the river between the dams, in Mm³/a (DWA, 2012; DWS, s.a.b).

Term	1987–2009	1987–1995	1996–2009
Inflow to Kromellenboog Dam (balance)	6.21	3.87	7.71
Passed on by Klein Maricopoort Dam (spill and release)	1.19	0.07	1.92
Net contribution of the river between the dams	5.01	3.80	5.80

6.3 Reliability of the canal supply

The water balance was tested before it was used, because the inflow is calculated from the same records (Table 9; Figure 9). The restriction rule was fitted on 1966–1974 and 1981–1986 and tested on 1987–2009 (the forward test), and fitted on 1987–2009 and tested on the earlier years (the reverse test). In the forward test the simulated storage and the yearly canal volume

meet all three criteria, and the monthly canal volume does not. In the reverse test the monthly canal volume meets the criteria and the yearly canal volume narrowly does not. The yearly canal volume is too high in the forward test and too low in the reverse test.

In each test the rule was fitted on the other period: 1966–1974 and 1981–1986 for the forward test, and 1987–2009 for the reverse test. The criteria are NSE above 0.5, RSR at most 0.7 and PBIAS within ±25% (Moriasi et al., 2007:891), applied here by analogy. A positive PBIAS means that the model is too low (Moriasi et al., 2007:888). The storage is not compared in the reverse test, because the area curve comes from the survey of 1982. The seasonal pattern is that of Figure 3 in the forward test and that of the years tested in the reverse test.

Table 9: Test of the water balance in both directions, with the rule fitted on the other period.

Measure	Forward test, 1987–2009: monthly storage	Forward: monthly canal volume	Forward: yearly canal volume	Reverse test, 1966–1974 and 1981–1986: monthly canal volume	Reverse: yearly canal volume
Values compared	276 months	265 months	20 years	178 months	14 years
NSE	0.86	0.38	0.60	0.56	0.49
RSR	0.37	0.79	0.64	0.66	0.72
PBIAS (%)	+7.4	−17.2	−22.4	+19.3	+19.6
Meets all three criteria	Yes	No	Yes	Yes	No

The reliability of the canal was then calculated against the registered volume over the 74 years (Table 10). With the rule, the canal receives the registered volume in 59% of the months and in 45% of the years, and without restriction in 87% and 77%. With the rule the canal fails about three times as often, but a failure month still brings half of the registered volume, while without restriction the canal then receives an eighth.

With the rule, the drought forms one failure event from December 1983 to January 1996, the longest of the record. Its shortfall is more than nine times the registered yearly volume, and it includes 40% of all failure months (Figure 10). Without restriction, the same drought breaks into separate events of no more than 20 months.

The rule costs water. The canal receives 0.21 Mm³/a less on average, and the dam loses 0.21 Mm³/a more to evaporation and spill. The reliability also depends on the period used to fit the rule: fitted on the fit period alone, the rule gives a reliability by time of 0.59, and fitted on the test period alone it gives 0.39 and a longest event of 169 months.

The adjusted inflow series (series 2) gives almost the same reliability, 0.588 with the rule and 0.865 without. The other inputs matter far less. With the rule, a change of 10% in evaporation, or halving or raising by half the release to the river, moves the reliability by time by about 0.02, and every other input, including a capacity 10% smaller and the seasonal pattern of demand, moves it by less than 0.01. Without restriction the largest change, also about 0.02, comes from a higher lowest usable storage. Leaving out the doubtful years of 1975–1980 lowers the reliabilities by at most 0.05.

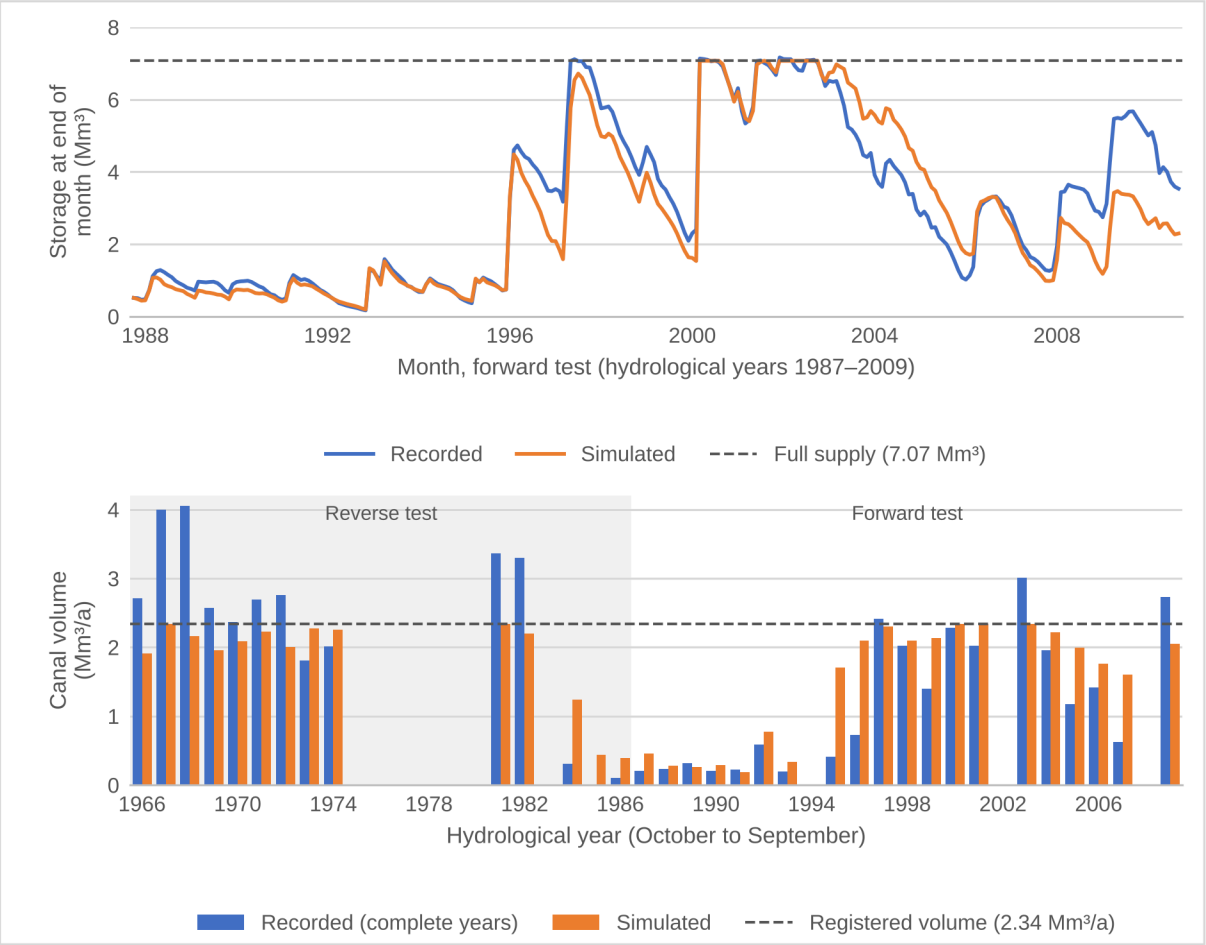


Figure 9: Upper panel: simulated and recorded monthly storage in the forward test, 1987–2009 (recorded storage from DWA, 2012). Lower panel: recorded and simulated yearly canal volume in the reverse test (1966–1974 and 1981–1986) and the forward test (1987–2009), against the registered 2.34 Mm³/a (DWS station A3H031).

The historic firm yield is the largest demand that the dam could have met in every month of the record without restriction. It is about 0.36 Mm³/a (Figure 11). One month fails at 0.37 Mm³/a, in October 1992, when the dam held 0.2 Mm³. At 1.5 Mm³/a the unrestricted canal fails in 8% of the months. The rule was fitted at the registered volume, so at lower demands the curve of the rule in Figure 11 is the rule as fitted.

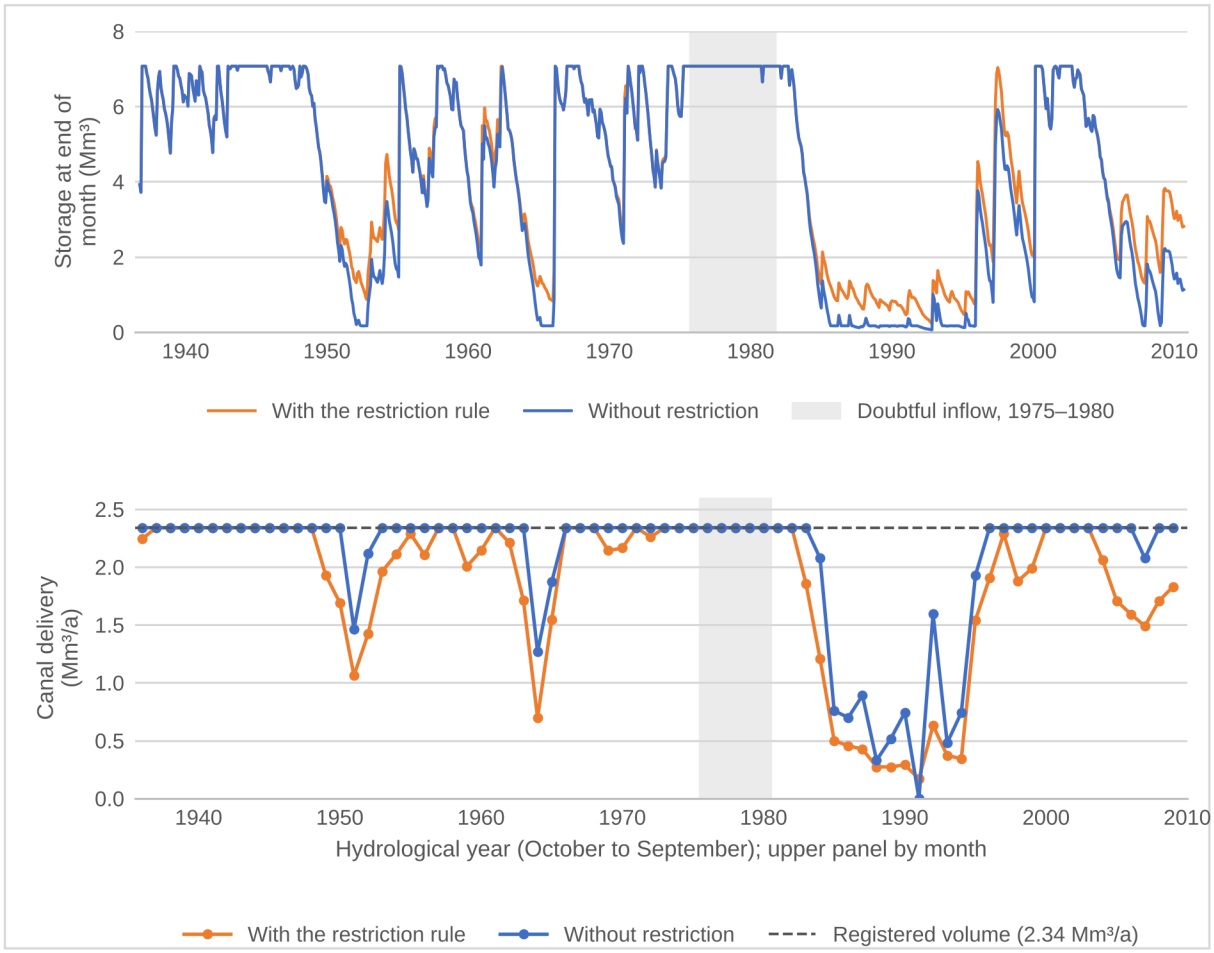


Figure 10: Reference run of the water balance, 1936–2009, with the restriction rule and without restriction. Upper panel: simulated monthly storage; lower panel: yearly canal delivery against the registered 2.34 Mm³/a. Shaded: 1975–1980, in which the inflow is doubtful.

Table 10: Reliability, resilience and vulnerability of the canal supply against the registered volume of 2.34 Mm³/a, Reference run with the inflow as recorded (series 1), hydrological years 1936–2009.

Criterion	With the restriction rule	Without restriction
Reliability by time	0.590	0.867
Reliability by volume	0.795	0.883
Yearly reliability	0.446	0.770
Shortfall events; longest (months)	16; 146	11; 20
Resilience	0.044	0.093
Vulnerability	0.50	0.88
Mean shortfall (Mm³/a)	0.48	0.27

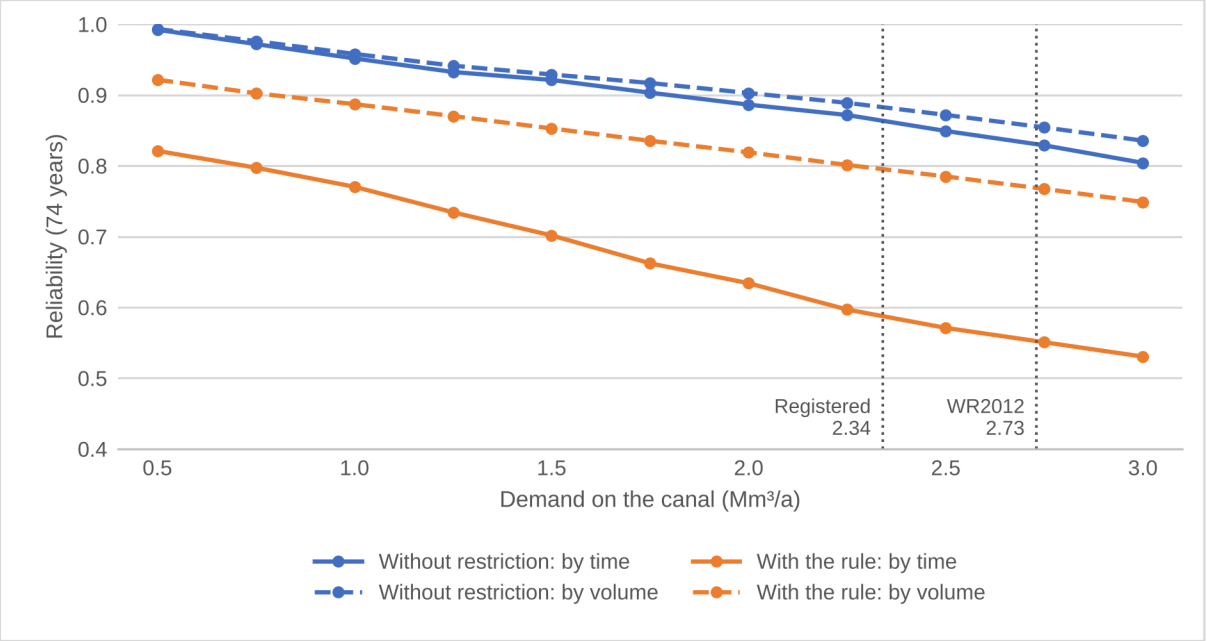


Figure 11: Reliability of the canal supply by time and by volume against the demand on the canal, with the restriction rule and without restriction, 1936–2009. Vertical lines: the registered volume (2.34 Mm³/a) and the demand of the WR2012 study (2.73 Mm³/a).

6.4 Fit and water budget of the groundwater model

The groundwater model was calibrated separately for the two structures on four static water levels, which gave different conductivities (Table 11). Structure S matches three of the four levels within 6 m, but it misses borehole 2526AD00180 in the valley by 28.9 m. In structure D both conductivities lie at the lower limits of their ranges, and the model fits the four levels through the depth of its ground drain. The transmissivity of the bedrock is far below the regional value in both structures.

The 262 archive levels that were not used in the calibration were then compared with both structures. Structure D has twice as many levels within 10 m as structure S, but it fits slightly worse than the benchmark (Table 11; Figure 12). On the lowest ground the measured water table is deeper than that of both structures, by twice as much in structure S. Structure S places the water table within 5 m of the surface at many more boreholes than the nine where the levels are that shallow. Under the irrigated land the depth to water is 4.5 m in structure S and 15.3 m in structure D (Map 7).

Grey marks levels whose status is not recorded, open circles the mean of each location group and blue the four static levels.

The benchmark is a water table 15.27 m below the ground everywhere. The lowest ground is the 28 of the 266 levels (the 262 and the four calibration levels) at 1 000–1 100 m. The budget closes to 0.00% in both structures; sums are up to 0.01 apart through rounding.

The water budget shows where the recharge goes (Table 11). The two structures receive the same recharge but lose it by different routes. The discharge to the rivers and streams in structure S exceeds the baseflow of 2.00 Mm³/a in the Reserve determination (DWS, 2021) and

the net contribution of the river between the dams (Table 8). In structure D it is below the baseflow. The comparison is of scale only, as the quantities are of different kinds.

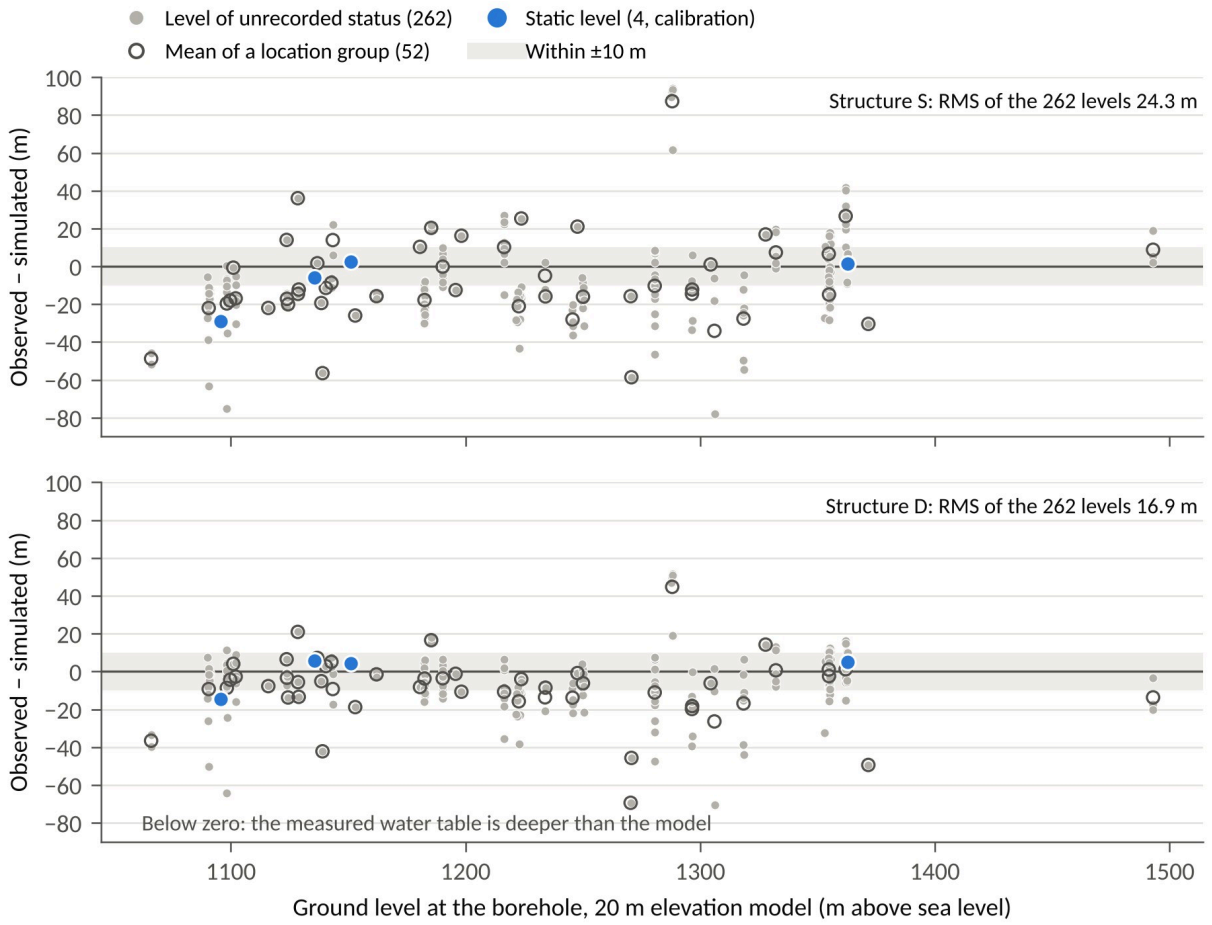
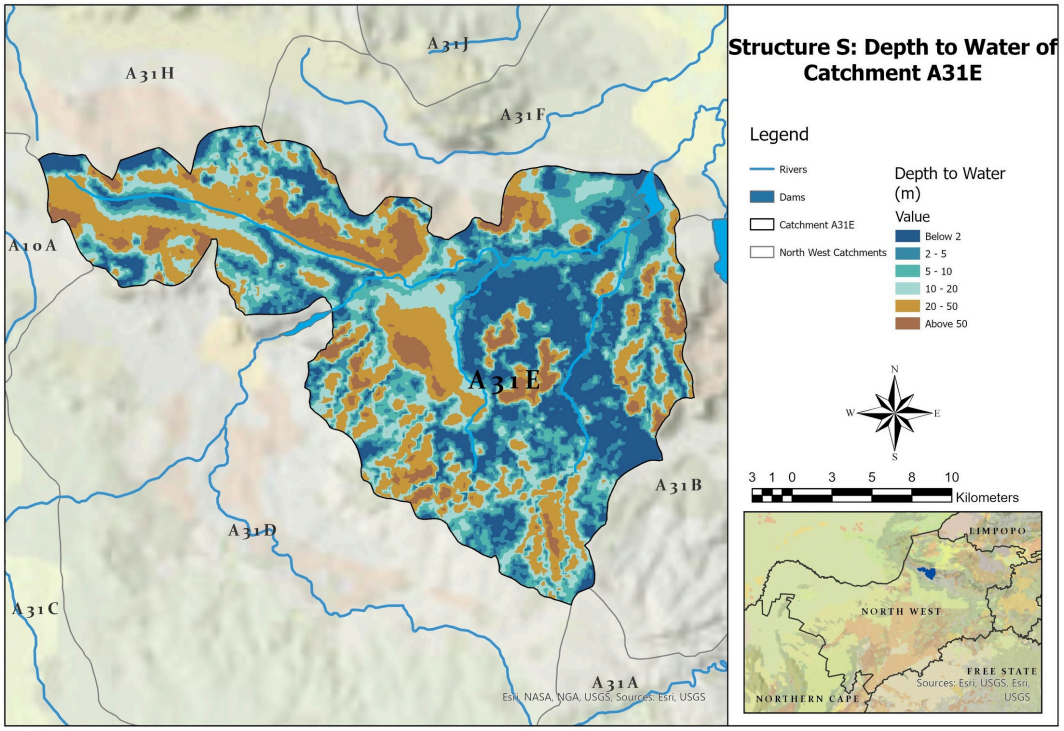


Figure 12: Residual (observed minus simulated head) at the single water levels recorded inside A31E (National Groundwater Archive, 1910–1994) against ground level, for structure S (upper panel) and structure D (lower panel).

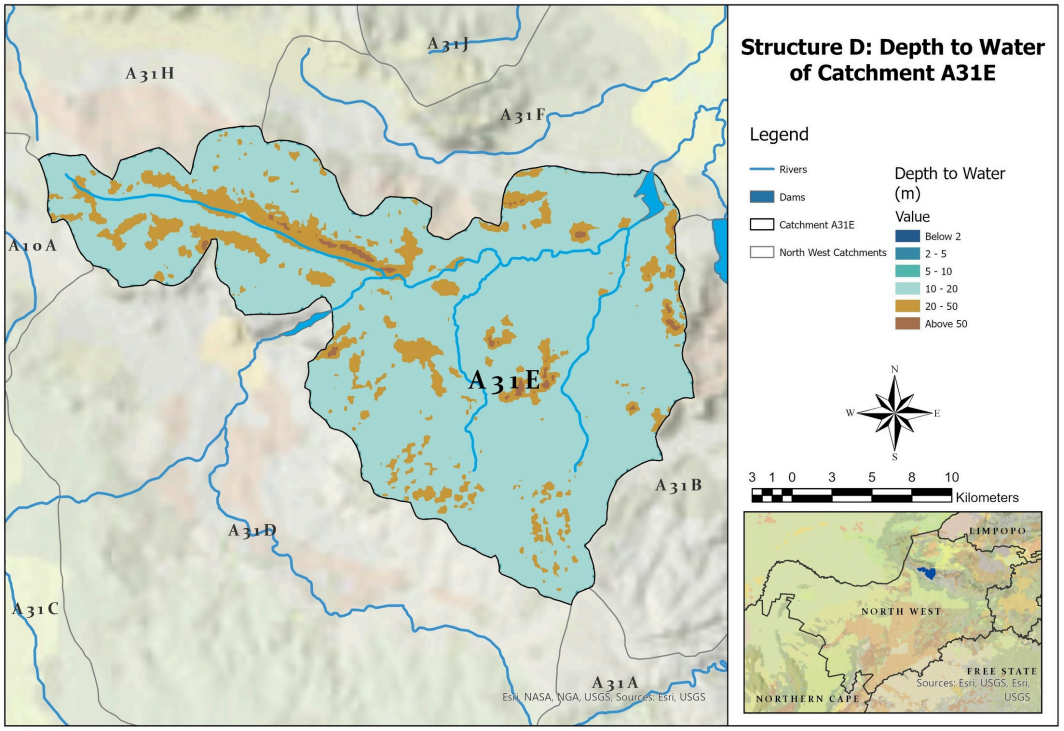
Table 11: Fit and water budget of the two base cases of the groundwater model, without wells.

Measure	Structure S	Structure D	Benchmark
Conductivity, surface deposits / bedrock (m/d)	0.430 / 0.0029	0.03 / 0.0005 (lower limits)	–
Transmissivity of the bedrock (m²/d; regional value 22.81)	0.60	0.10	–
Four calibration levels: root mean square error (m)	14.85	8.53	8.61
Same, without borehole 2526AD00180 (m)	3.85	5.03	5.15
262 other levels: root mean square error (m)	24.3	16.9	14.9
262 other levels: within 10 m (%)	31	62	71
Lowest ground (28 of 266 levels): mean residual (m)	−23.1	−11.0	−11.8
Boreholes where the water table is within 5 m of the surface (measured: 9 of 262)	114	14	–
Recharge (Mm³/a)	17.53	17.53	–
Out: ground drain (Mm³/a)	7.67 (44%)	16.90 (96%)	–
Out: rivers and streams (Mm³/a)	9.86 (56%)	0.63 (4%)	–

(a) Structure S



(b) Structure D



Map 7: Simulated depth to water in the groundwater model of A31E for (a) structure S (outflow 1 m below the surface) and (b) structure D (outflow 15.27 m below the surface), with the rivers and streams and the two dams.

The Reserve determination gives a recharge of 17.336 Mm³/a, a groundwater use of 0.81 Mm³/a and a baseflow of 2.00 Mm³/a (DWS, 2021). This leaves 14.5 Mm³/a of the recharge unaccounted for. The recharge is almost nine times the baseflow. The recharge of the model, 17.53 Mm³/a, is slightly more.

6.5 Groundwater abstraction to offset the shortfall

Abstraction was added under the irrigated land to see how far groundwater could offset the shortfall and where the pumped water would come from. In structure S the mean drawdown in the 45 pumped cells is 2.30 m at 0.5 Mm³/a and 40.8 m at the registered volume of 2.34 Mm³/a (Figure 13). In structure D the drawdown is five and nine times larger at 0.25 and 0.5 Mm³/a, and from 1.0 Mm³/a the pumped cells deliver at most about 1.2 Mm³/a.

Open circles in the figure mark rates that structure D cannot deliver. The shaded bands show the mean shortfall against the registered volume in the Reference run and cases M and D, for both bounds (0.27–0.70 Mm³/a), and the rate during the longest shortfall event of the Reference run (1.80–2.05 Mm³/a).

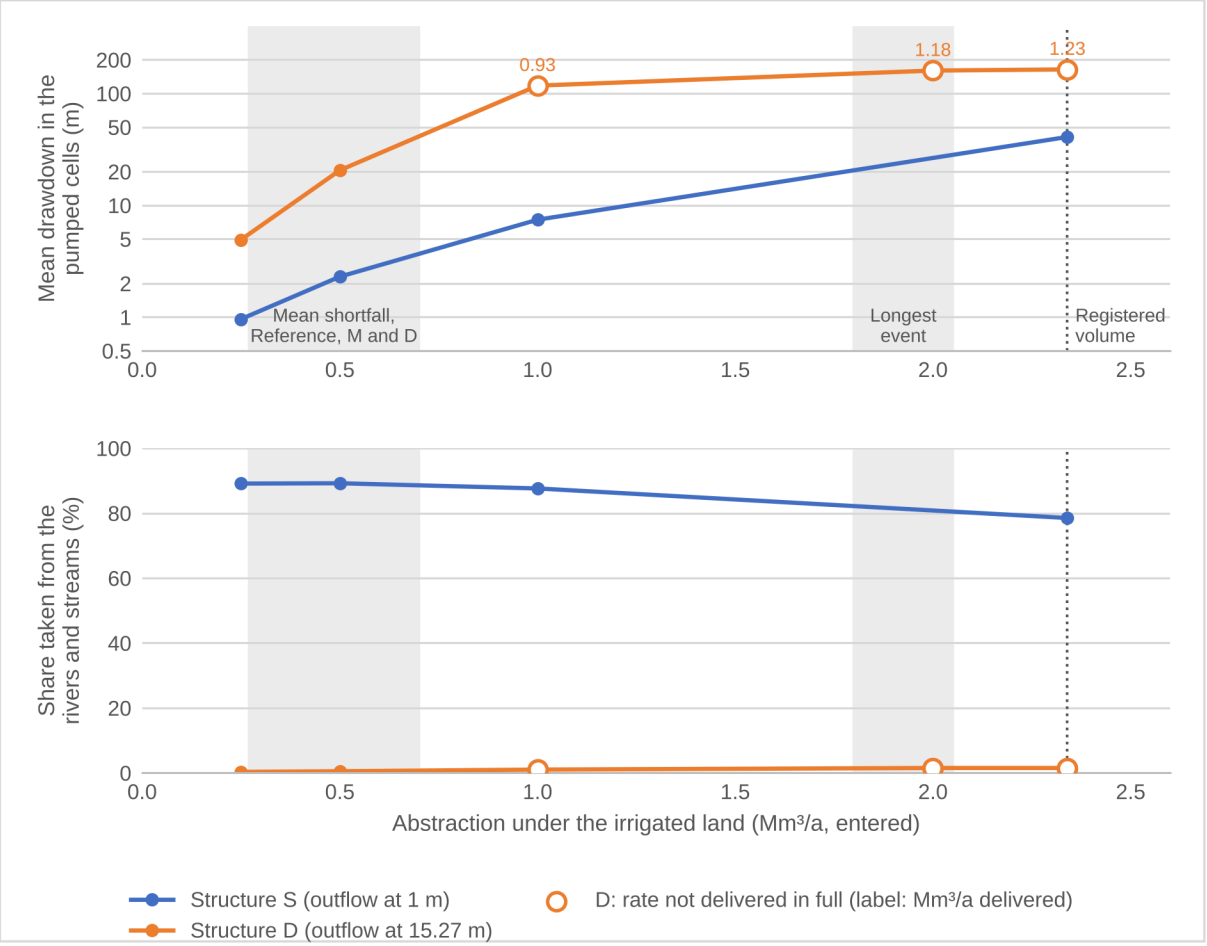


Figure 13: Steady-state response of the groundwater model to abstraction spread over the 45 model cells under the irrigated land, for structure S and structure D. Upper panel: mean drawdown in the pumped cells; lower panel: share of the abstraction balanced by less discharge to the rivers and streams.

The source of the pumped water is opposite in the two structures. In structure S, 88% to 89% of the pumped water comes from less discharge to the rivers and streams at rates up to 1.0 Mm³/a, and more than three-quarters at the registered volume. At 0.5 Mm³/a, about 69% comes from the Klein Marico River itself and 20% from the lower Waterkloofspruit. In structure D, the rivers and streams supply under 2%, and the rest comes from the ground drain.

Further cases each changed one part of the set-up (Table 12). With the conductivities of structure S, the drawdown in structure D falls to about that of structure S, but its water still comes entirely from the ground drain. Spreading the abstraction over the surface deposits within 1 km reduces the mean drawdown to a quarter in structure S and to one twenty-eighth in structure D, without changing the source. Abstraction from the bedrock within 1 km lowers the share from the rivers and streams in structure S to 58%. The halved and doubled models fit the four water levels equally well, and their drawdowns in structure S differ by a factor of about eight.

The means are over the 45 cells of P1, the 474 cells of P2 and the 410 cells of P3. The halved and doubled models give the same water levels as the calibrated model at 1.0 and 0.25 Mm³/a, so their values equal those runs.

Table 12: Mean drawdown in the pumped cells, and share of the abstraction that comes from less discharge to the rivers and streams, for 0.5 Mm³/a in cases that each change one part of the set-up.

Case	S: drawdown (m)	S: from rivers and streams	D: drawdown (m)	D: from rivers and streams
P1, under the irrigated land	2.30	89.2%	20.6	0.5%
P2, spread over the deposits within 1 km	0.55	88.8%	0.73	2.7%
P3, on the bedrock within 1 km	3.93	58.0%	Not run	Not run
P1, conductivities of S in structure D	–	–	2.22	0%
P1, recharge and conductivities halved	7.43	87.6%	117.5 (93% delivered)	1.0%
P1, recharge and conductivities doubled	0.96	89.1%	4.92	0.3%

The shortfall was then set against the recharge. The mean shortfall of the Reference run (Table 10) is under 3% of the recharge of the Reserve determination, and the registered volume is 13.5% of it. The rate needed during the longest shortfall event, 1.8 to 2.05 Mm³/a, cannot be delivered from under the irrigated land in structure D at the regional recharge. In structure S it lies between the runs at 1.0 and 2.34 Mm³/a.

6.6 Climate change

The two climate cases of Table 6 were applied to the water balance of the dam and to the groundwater model, to test how the canal supply and the groundwater respond in the two sensitivity cases for about 2050. Case M is the median step and case D the dry end of the projections. The reliability of the canal supply is measured against the registered volume raised by the demand factor of the case (Section 5.4.4). Table 13 gives the results against the Reference run.

The reliability by time falls in each step, with and without the rule, and the mean shortfall grows. The longest event grows from the Reference run to case M, with and without the rule, and with the rule it grows further in case D. With the rule, the reliability by time falls from 0.59 in the Reference run to 0.54 in case M and 0.48 in case D. The dam delivers about as much water in case M as in the Reference run and less in case D (1.79 against 1.86 Mm³/a with the rule).

Canal supply is given with the restriction rule / without restriction, for hydrological years 1936–2009, against the registered volume raised by the demand factor of the case. Groundwater is the discharge to rivers and streams in the base case (Mm³/a) and its change under the recharge factor of the case.

Table 13: The climate cases against the Reference run.

Case	Reliability by time	Yearly reliability	Mean shortfall (Mm³/a)	Longest event (months)	Groundwater to rivers, S	Groundwater to rivers, D
Reference	0.590 / 0.867	0.446 / 0.770	0.48 / 0.27	146 / 20	9.86 Mm³/a	0.63 Mm³/a
M (median)	0.541 / 0.829	0.405 / 0.703	0.60 / 0.36	148 / 35	−3.8%	−5%
D (dry end)	0.477 / 0.791	0.324 / 0.635	0.70 / 0.44	163 / 35	−10.5%	−14%

The simulated storage is never higher than in the Reference run, and it is lower at every level in Figure 14.

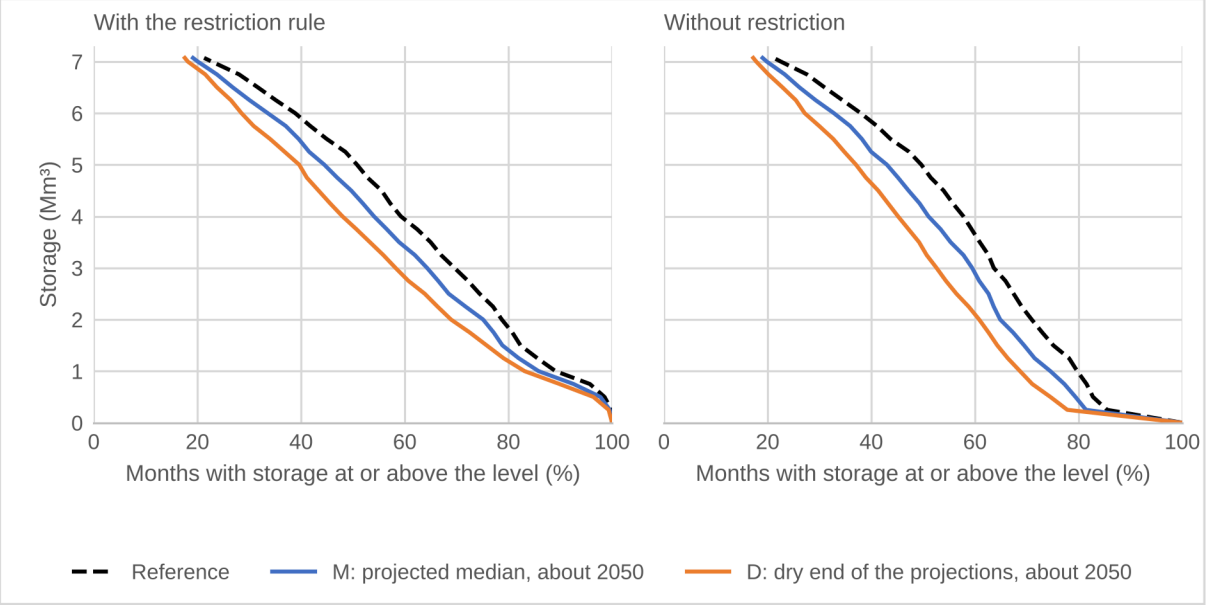
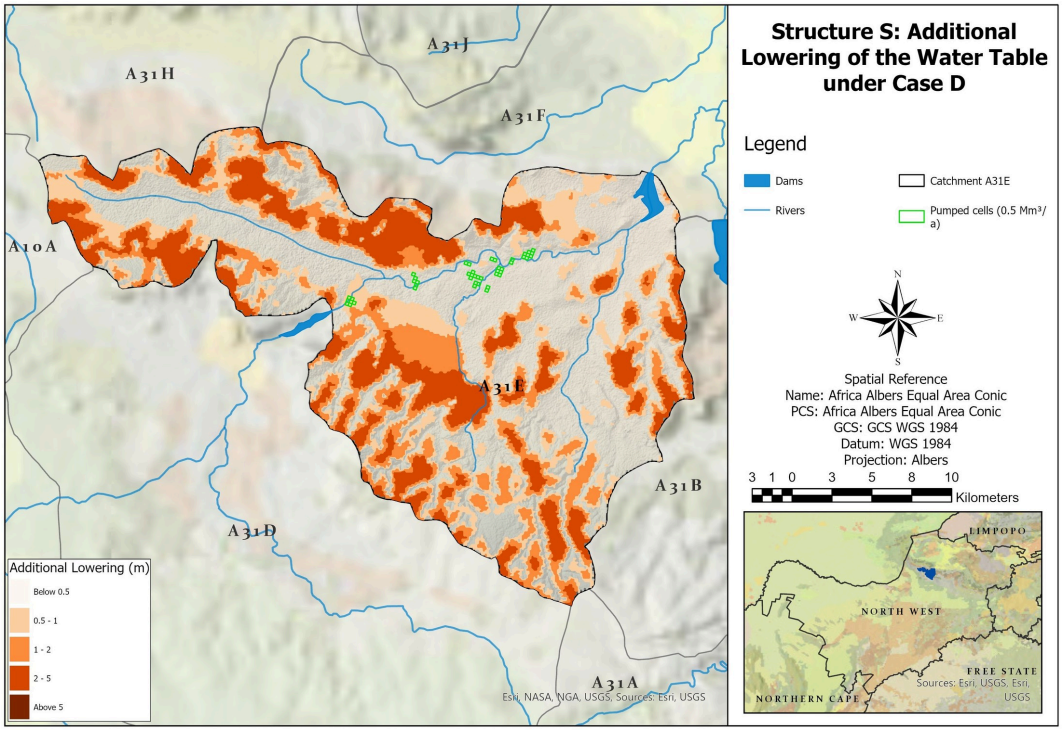


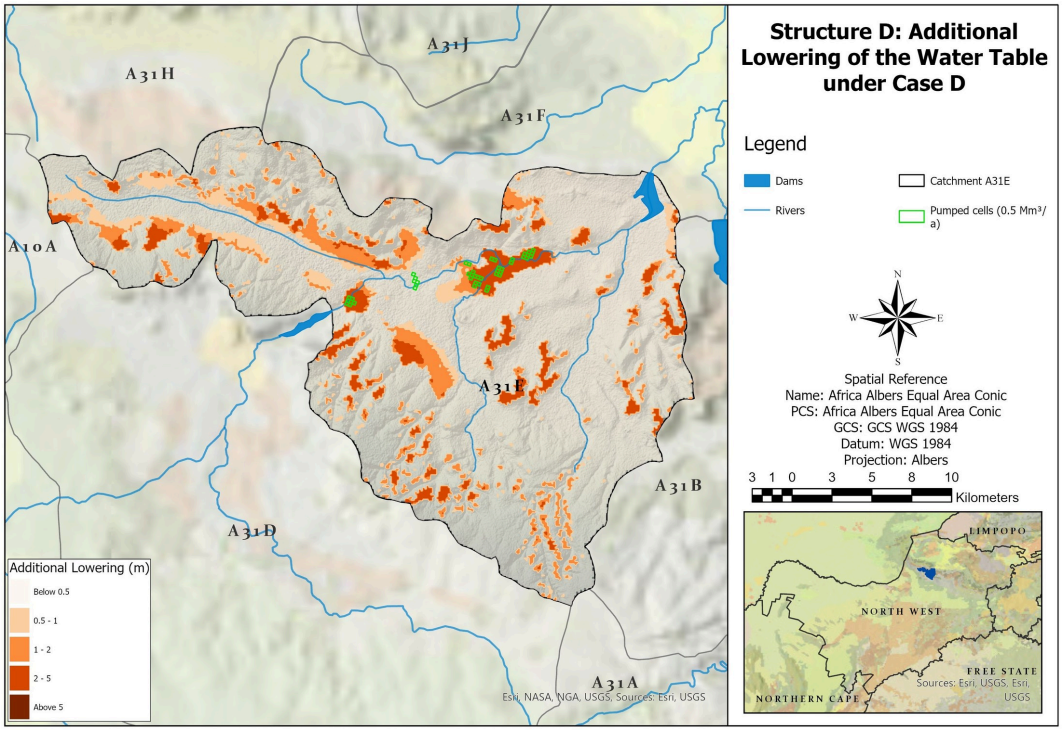
Figure 14: Share of months in which the simulated storage of Klein Maricopoort Dam is at or above each level, for the Reference run and cases M and D, 1936–2009. Left: with the restriction rule; right: without restriction.

In the groundwater model the discharge to the rivers and streams falls by about the same percentage as the recharge: by 3.8% and 10.5% in structure S and by 5% and 14% in structure D (Table 13). Map 8 shows how much lower the water table stands in case D than in the Reference run, from two runs that both include an abstraction of 0.5 Mm³/a under the irrigated land. The mean lowering is 1.00 m in structure S and 0.38 m in structure D. Under the pumped cells it averages 0.23 m in structure S and 3.78 m in structure D, where it about equals the increase in the drawdown of the abstraction, from 20.6 to 24.3 m. Under case M the lowering is about a third of that under case D.

(a) Structure S



(b) Structure D



Map 8: Additional lowering of the simulated water table when recharge falls to 89% of the base value (case D), for (a) structure S and (b) structure D, with 0.5 Mm³/a abstracted from the 45 model cells under the land mapped as irrigated in both runs.

6.7 Flood extent and depth in the modelled reach

The design flood at F2 was estimated with the SDF method (Section 5.2.3), and two model runs were routed through the 15.7 km reach, to find how large the floods are and how much of the irrigated land they cover.

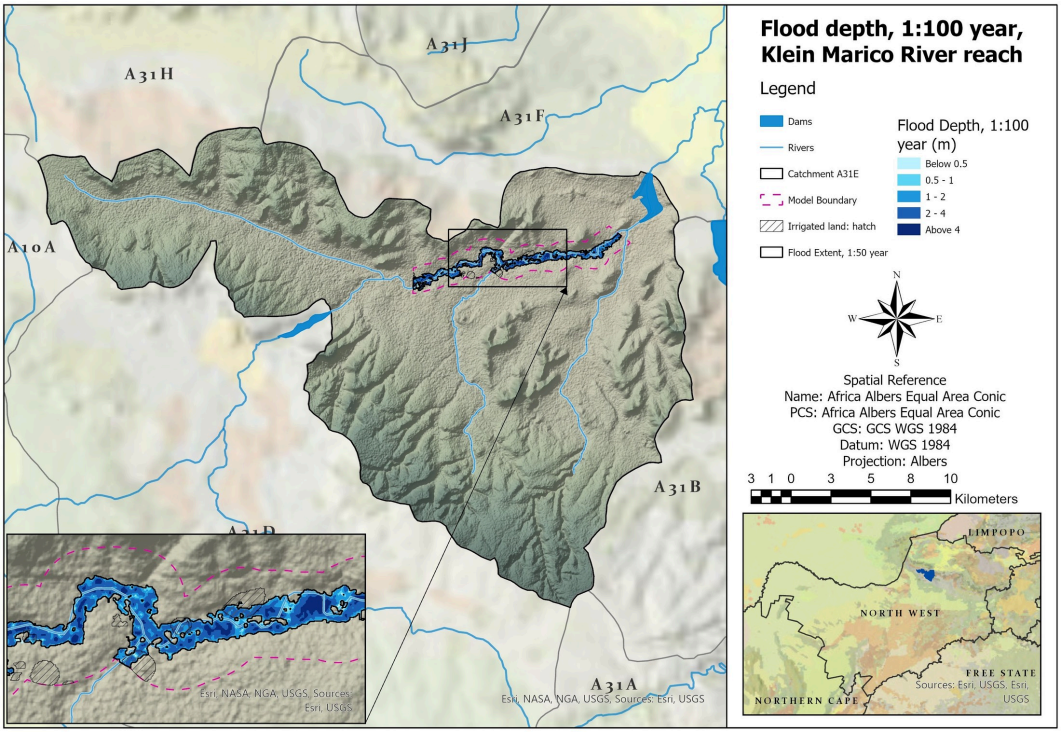
The catchment above the reach is underlain for 46.4% by dolomite of the Malmani Subgroup, all of which lies above Klein Maricopoort Dam, while A31E itself contains none (Michaluk & Keyser, 1991; Walraven, 1981). At the 1:50-year and 1:100-year return periods the SDF gives peaks of 974 and 1 252 m³/s (Table 14).

Table 14: Design flood peaks at F2 and the floods routed through the modelled reach (SDF method; catchment about 1 357 km²; model area 24.3 km²; irrigated land in the strip below the dam 165 ha, DFFE, 2024).

Run	Design peak (m³/s)	Peak in / out (m³/s)	Peak reduced	Flooded (km²)	Deeper than 1 m (km²)	Irrigated land flooded (ha)	Of which deeper than 1 m (ha)
1:50-year	974	972 / 951	2.1%	7.72	6.26	42.8	35.1
1:100-year	1 252	1 250 / 1 236	1.1%	7.97	6.66	44.2	38.1

Over the reach the peak decreases by 1% to 2% and arrives about two hours later. The flooded area changes much less than the peak. The 1:100-year peak is 29% larger than the 1:50-year peak, but it floods 3% more land. The outlines of the two floods almost coincide on Map 9. The 1:100-year flood covers 0.25 km² more land, equal to a strip about 8 m wide along each side of the 15.7 km reach.

About a quarter of the irrigated land in the strip is flooded in each run, and most of the flooded part lies under more than 1 m of water.



Map 9: Maximum depth of the 1:100-year flood (SDF method) in the modelled reach of the Klein Marico River, from the HEC-RAS 2D model, with the outline of the 1:50-year flood. Hatching shows land mapped as irrigated; the inset shows the irrigated land in more detail. Depth classes 0.01–0.5, 0.5–1, 1–2, 2–4 and >4 m. The depths were exported from HEC-RAS Mapper and drawn in ArcGIS.

6.8 Contaminant pathways and water quality

Two things were examined. Particles were released from the 45 model cells under the irrigated land in the runs without abstraction and tracked forward in both model structures (MODPATH; Pollock, 2016), to see where water that recharges under the fields goes. The records of the Department of Water and Sanitation (DWS) were screened against the guidelines of DWAF (1996a, 1996b) to compare the water of the river below Klein Maricopoort Dam with the water of A31E, which is the Wilgeboomspruit and the seven groundwater samples.

In structure S, 387 of the 405 particles (96%) end in a cell where a river drain discharges (Map 10). The median travel time is about 140 years at an effective porosity of 0.10, and it scales with the porosity. In structure D no particle reaches a river or stream, and 288 of the 405 do not leave the cell in which they started.

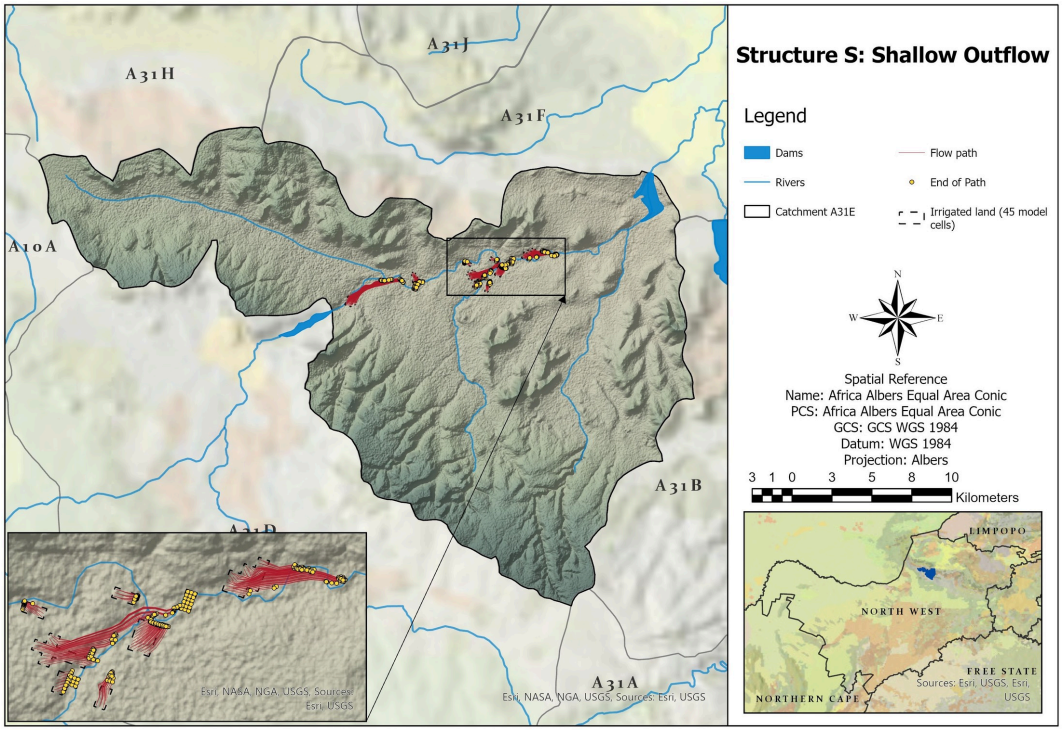
The water-quality records are in Figure 15 and Table 15. Below Klein Maricopoort Dam the median electrical conductivity (EC) was 51.2 mS/m in 1989–1993 (station A3H030, 56 samples) and 62.3 mS/m in 2012–2023 (A3H042Q01, 81 samples), and 86% and 99% of the samples exceeded the upper limit of the target range for irrigation, 40 mS/m. The three sites along the river between the dams (188257, 188038 and 188030, with three, three and one sample in 2004–2007) had medians of 76 to 83 mS/m.

The Wilgeboomspruit, the one gauged stream inside A31E (1971–1991), had a median of 3.85 mS/m and no exceedance, and the seven groundwater samples of A31E (DWS, 2021:163) a median of 17.7 mS/m. Kromellenboog Dam, which receives both waters, lies between them and the river below the dam at 21.7 mS/m (1966–2023). Sulphate is also higher along the river than in Kromellenboog Dam and the Wilgeboomspruit, with medians of 30 and 57 mg/L below the dam and 102 to 145 mg/L between the dams, against 17.7 and 3.05 mg/L. Chloride, sodium and SAR are above their guideline values in none of the samples at any site. The chloride and sodium values are those for water that wets the foliage, and SAR is screened against 2.0, the value for the crop yield (DWAF, 1996a:134). Phosphate-P exceeds 0.025 mg/L, the boundary between the mesotrophic and eutrophic classes (DWAF, 1996b:97), in 19% of the samples at A3H042Q01 (74 samples, 2012–2023) and 4% at A3H030 (56 samples, 1989–1993) below the dam, 29% in Kromellenboog Dam and 20% in the Wilgeboomspruit, so the nutrients do not separate the two waters in the same way.

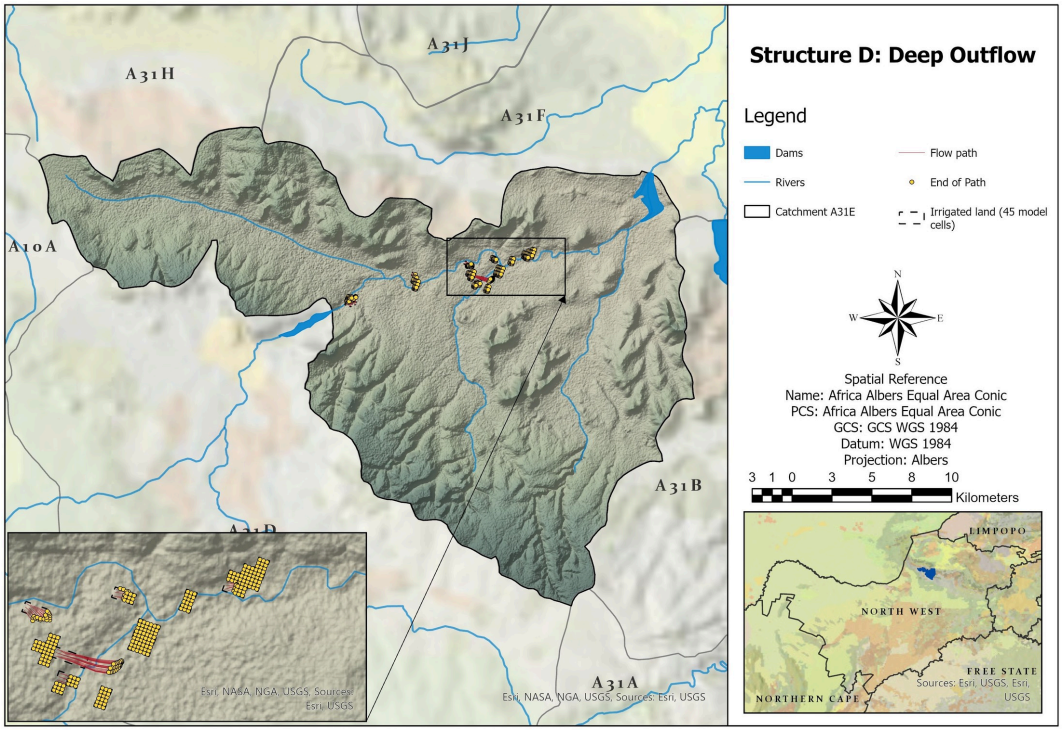
The figure shows the upper limit of the target range for irrigation (40 mS/m; DWAF, 1996a:155) and the median EC of the groundwater of A31E (DWS, 2021). Single samples are shown as circles.

Table 15 gives, for each site, the percentage of the samples analysed for a quantity that exceed the guideline value. The guidelines are EC above 40 mS/m, the upper limit of the target range for irrigation (DWAF, 1996a:155); inorganic nitrogen above 0.5 and 2.5 mg/L and phosphate as phosphorus above 0.025 mg/L, the boundaries of the trophic classes, which the guideline defines for the average summer concentration (DWAF, 1996b:84, 97). Chloride (100 mg/L) and sodium (70 mg/L), the values for water that wets the foliage, and SAR (2.0) exceed their guidelines in none of the samples at any site (DWAF, 1996a:48, 134, 144). Sulphate has no guideline in this screening.

(a) Structure S



(b) Structure D



Map 10: Flow paths of 405 particles released in the 45 model cells under the irrigated land (MODPATH 7; Pollock, 2016), for (a) structure S and (b) structure D. Yellow dots mark the end of each path. In S, 387 of the 405 paths end at a river; in D none do. The paths show advective flow direction only, and travel times depend on an assumed porosity.

The DRASTIC index was taken from the national map to rate how vulnerable the aquifer is to contamination from the surface. It is low over 96.7% of A31E and moderate over 3.3% (Map 11; DWAF, cited by Esterhuyse, 2017).

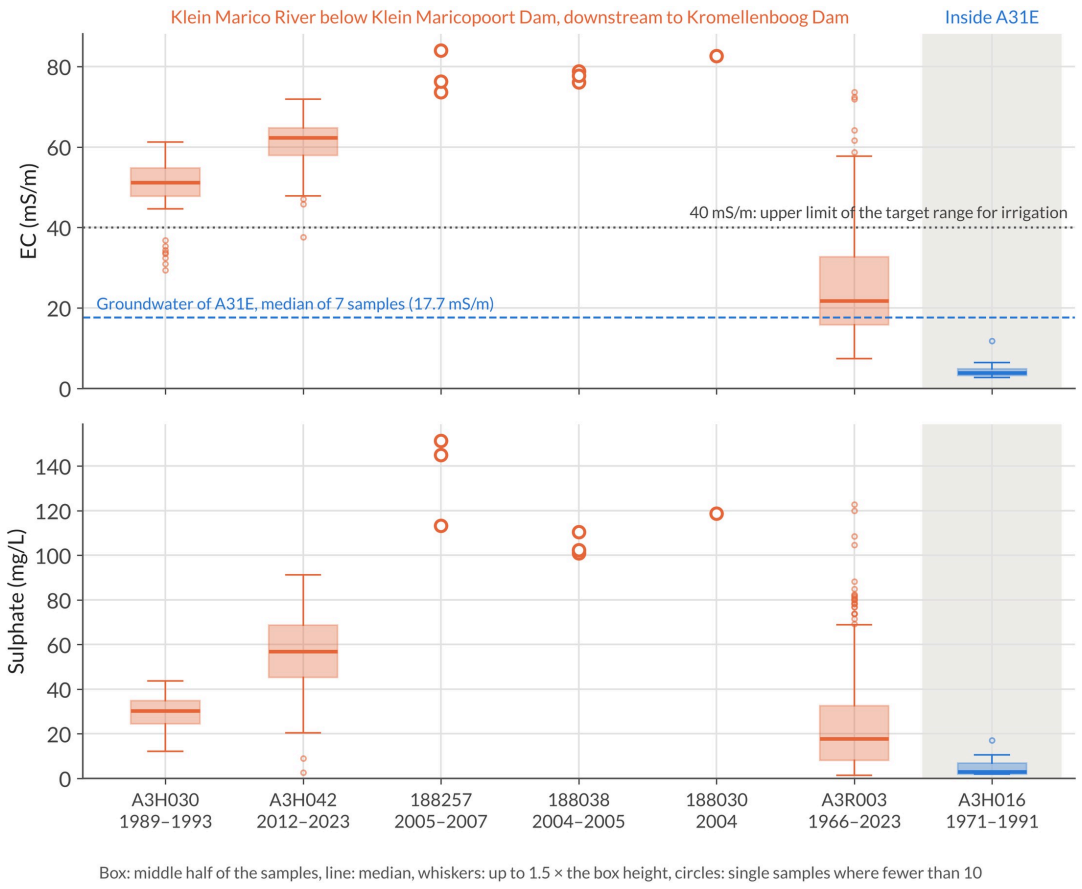
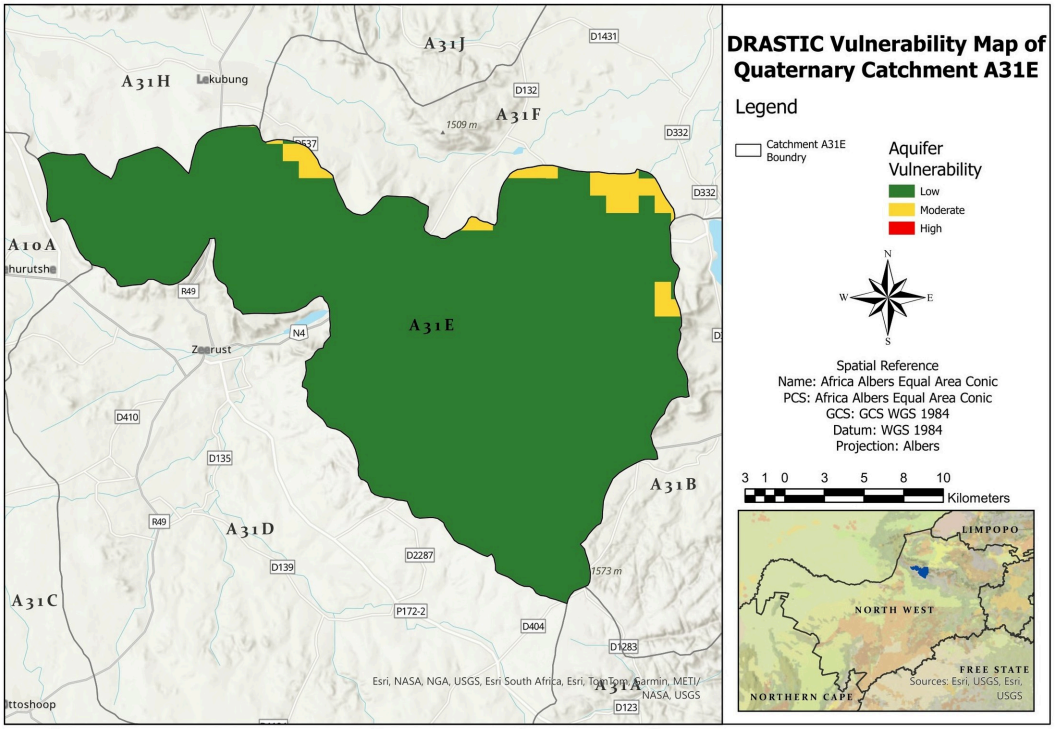


Figure 15: Electrical conductivity (upper panel) and sulphate (lower panel) at the DWS water-quality sites along the Klein Marico River below Klein Maricopoort Dam, in Kromellenboog Dam and in the Wilgeboomspruit (DWS, s.a.e).

Table 15: Screening of the DWS water-quality records along the two paths (DWS, s.a.e; surface samples).

Site	Path	Period	Samples	EC median (mS/m)	EC above guideline	Inorganic N above 0.5 / 2.5 mg/L	Phosphate-P above 0.025 mg/L
A3H030	1	1989–1993	56	51.2	86%	2% / 0%	4%
A3H042Q01	1	2012–2023	81	62.3	99%	12% / 4%	19%
188257	1	2005–2007	3	76.3	100%	0% / 0%	0%
188038	1	2004–2005	3	77.7	100%	0% / 0%	33%
188030	1	2004	1	82.6	100%	100% / 0%	100%
A3R003Q01	1 and 2	1966–2023	248	21.7	17%	16% / 0%	29%
A3H016Q01	2	1971–1991	54	3.85	0%	5% / 2%	20%



Map 11: Aquifer vulnerability of A31E according to the DRASTIC index (Aller et al., 1987), in the classes of the national map: low (below 90), moderate (90–140) and high (above 140) (DWAF, cited by Esterhuyse, 2017). The index is mapped on a 1 km national grid and is not a measurement.

7. Discussion

The canal supply from Klein Maricopoort Dam cannot be relied on for the registered volume. The dam is small in relation to that volume, so a drought of several years empties it, and the manner in which deliveries are restricted as it empties sets the balance between how often the supply fails and how severe a failure is (Hashimoto et al., 1982:18–19). Why the inflow remained low during the drought is not yet settled. The historic firm yield of the dam is also well below the 1.4 Mm^3/a that the Department gives, a figure for which the report states no method and which rests on hydrology that has not been updated since 1988 (DWAF, 2004:50, 54, 83).

The drought further shows that A31E carries two waters, which differ in origin and in quality. The water in the canal comes from outside the catchment; it fell short during the drought, and in most samples its salinity is above the target for irrigation. The Wilgeboomspruit and the groundwater of A31E are much fresher, but this water enters the river below the canal and therefore does not reach it.

Because groundwater is tied to both waters, it can back up the canal only in part. Water pumped from an aquifer often comes from less discharge elsewhere (Leake & Barlow, 2013). In structure S it comes mostly from the Klein Marico River, and so from the water that feeds Kromellenboog Dam. Which structure is closer to the catchment the records cannot yet show: the 262 other water levels favour structure D on the share within 10 m, but neither structure fits them better than a constant depth to water. The same question decides where water from the irrigated fields goes, since in structure S nearly all of it would reach a river, whereas in structure D none would.

Climate change adds to these limits. In the two sensitivity cases for about 2050, the canal supply is less reliable than in the Reference run, most of all in the dry case, and in both structures the discharge of groundwater to the rivers and streams falls by about as much as the recharge. Planning should therefore be tested against the dry end of the projections rather than against the median.

For part of the irrigated land, shortage is not the only risk. This land lies on the floodplain, so its users face both a shortage that can last for years and flooding. The model shows that about a quarter of the irrigated land in the strip is covered by the 1:100-year flood.

Taken together, planning in the catchment should assume droughts of several years rather than the registered volume, and groundwater cannot yet be relied on as a backup, because its effect on the river has not been established. Each remaining range has a cause that data could narrow: for the canal, the restriction rule, and for groundwater, the position of the water table relative to the river bed.

8. Recommendations

Each recommendation follows from the Discussion (Section 7) and names its addressees.

1. **Plan for shortfalls that last several years (DWS; Premier's Office).** Drought plans should treat the registered volume as a volume that the dam cannot assure, and should be tested against the dry end of the climate projections and not the median.
2. **Review the yield that the Department gives for the dam (DWS).** The report that gives the yield states no method, and the Marico hydrology has not been updated since 1988. The historic firm yield found here is well below it.
3. **State the restriction rule of the scheme and make it known to the irrigators (DWS).** The records do not show which rule was applied, and the rule sets the balance between how often and how badly the canal fails.
4. **Do not rely on groundwater as a backup until its effect on the river has been assessed (DWS; Premier's Office; AGES).** In one structure of the model most of an abstraction comes from the river that supplies Kromellenboog Dam.
5. **Monitor the quality of the river below the dam, the inflow to the dam and boreholes below the irrigated land (DWS; AGES).** The river below the dam is above the salinity target for irrigation in most samples, and the source of its salt and the quality of the groundwater below the fields are not known.
6. **Collect the data that would reduce the uncertainty most.** The restriction rule and the position of the water table relative to the river bed decide most of the ranges, so the priorities are:

   (a) The operating and restriction rule of the scheme, with yearly records of the water ordered and delivered (DWS).

   (b) A gauged record of the inflow to Klein Maricopoort Dam and of the water used above it, to settle the cause of the low inflow (DWS).

   (c) Water levels and electrical conductivity in boreholes beside the Klein Marico River below the dam (DWS; AGES).

   (d) The lithological logs of the valley deposits in the archive, and pumping tests in those deposits (AGES).

   (e) Water-level series in the valley and in the western arm (AGES).

9. Conclusions

This study set out to quantify the reliability of the canal supply from Klein Maricopoort Dam, how far groundwater within A31E could offset its shortfalls and how both respond to a drier climate, and to screen the floods and the contaminant paths along the irrigated reach.

The canal supply is not reliable for the registered volume. The dam is small against it, and one drought of many years, in which the inflow stayed low, left the canal short year after year. Groundwater within A31E cannot yet be relied on as a backup. In one structure of the model most of the pumped water would come from the river that feeds Kromellenboog Dam, and the water levels available cannot show which structure is closer to the catchment. In the sensitivity cases for about 2050, the canal supply and the discharge of groundwater to the rivers are both lower, most in the dry case.

Part of the irrigated land lies on the floodplain and is covered by the 1:100-year flood. Water from under the irrigated land would reach a river in one structure and not in the other. The river below the dam is above the salinity target for irrigation in most samples, and the water of A31E is fresher.

The registered volume cannot be assured by the dam, and the recommendations follow from this: plan for droughts of several years, review the yield, state the restriction rule, and collect the data that would narrow the two main unknowns.

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AI Declaration Page

Title of the work: Storage, drought and climate change: the reliability of the canal supply from Klein Maricopoort Dam and the role of groundwater in A31E, North West Province.

Author: LH Stroebel (student number 42766613). Supervisor: Sascha Roopa.

1. Was AI used? Yes.

2. Purpose. AI was used to process data and build the dam water-balance spreadsheet from the author’s records, to draw the graphs, to draft and edit text, to structure the document and to check facts and references.

3. Tool used. Claude (Anthropic).

4. How AI was applied in the research process. The author supplied the records, defined the study and made the decisions on method. Claude processed the records and built the dam water-balance spreadsheet from them, drew the graphs, drafted and edited text and structured the document. The Introduction and the first version of the Study Area were written by the author, and Claude then edited the text and extended it to cover the whole project. The author ran the GMS/MODFLOW, MODPATH and HEC-RAS models, made the maps in ArcGIS Pro and checked the outputs against the records.

5. How the accuracy and reliability of the AI outputs were verified. The author checked the model outputs against the records. Claude was also used to check facts and references against the sources. The full log of the work with Claude is available on request.

Signature: _____

Date: _____